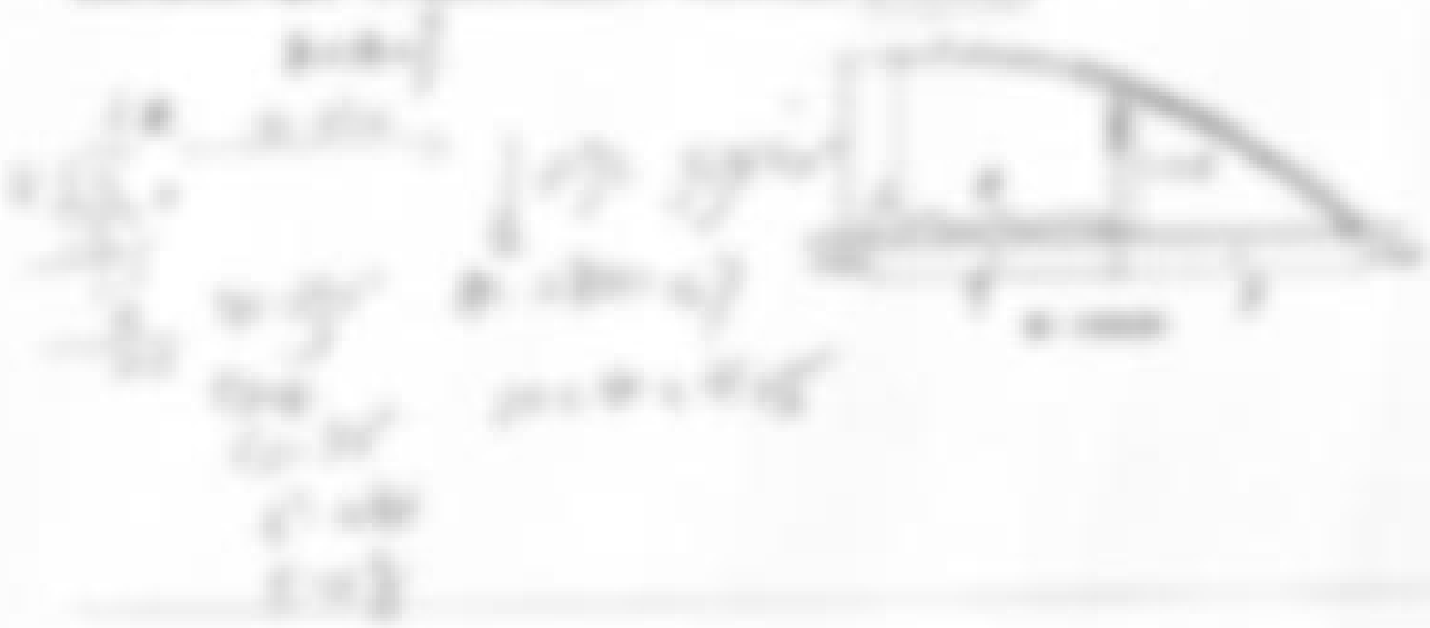


1. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle θ to the horizontal. The particle follows a parabolic path and returns to the horizontal axis at a distance R from O .

Answer: (10 marks)



2. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle θ to the horizontal. The particle follows a parabolic path and returns to the horizontal axis at a distance R from O .

Answer: (10 marks)

3. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle θ to the horizontal. The particle follows a parabolic path and returns to the horizontal axis at a distance R from O .



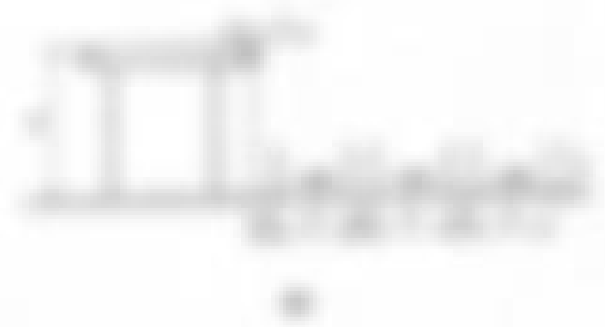
1. **QUESTION** (10 marks)
A beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

2. **ANSWER** (10 marks)



3. **QUESTION** (10 marks)
A beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

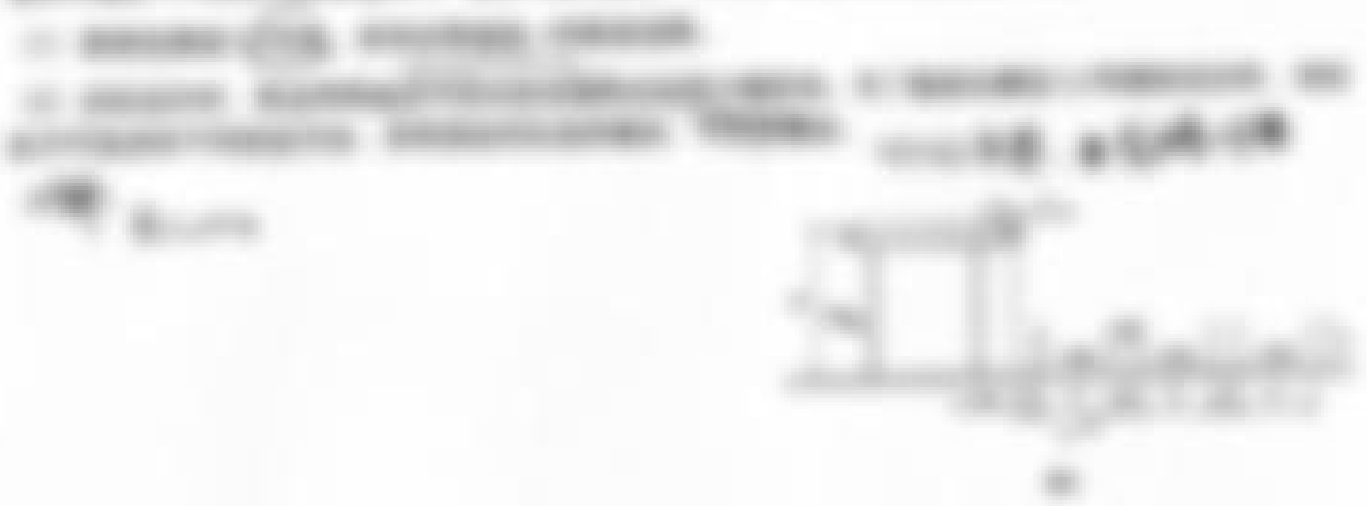
4. **ANSWER** (10 marks)



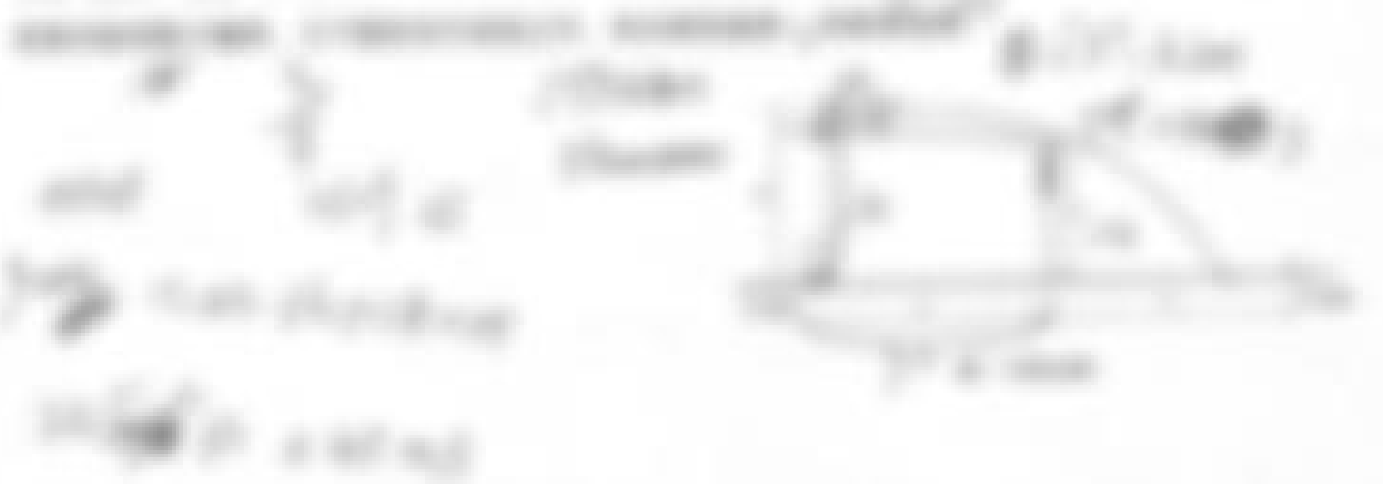
1. The following information was obtained from a survey of the population of a certain country in the year 2000. The population was 10 million. The number of people in each age group is given in the table below.



2. The following information was obtained from a survey of the population of a certain country in the year 2000. The population was 10 million. The number of people in each age group is given in the table below.



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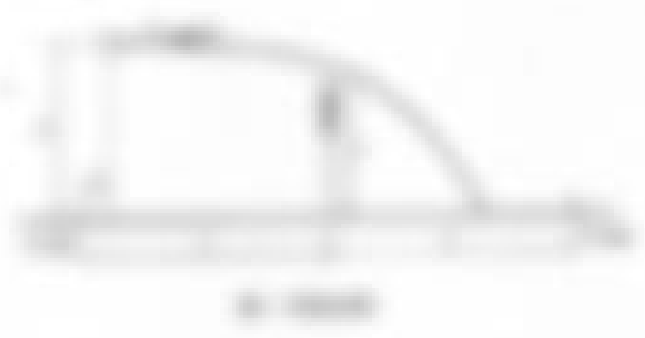
1. *Journal of Management Education*, 2000, 24(1), 1-10. doi:10.1177/0095647200241001

1. **QUESTION** **ANSWER** **QUESTION** **ANSWER**

1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top edge of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .

Diagram: A rectangular block of height h and width b is shown. A force F is applied to the top-left corner, directed at an angle θ to the horizontal. The block is on the verge of tipping over the right edge.

Given: m, h, b, θ



2. A uniform rod of length L and mass M is pivoted at the left end. A force F is applied to the right end, directed at an angle θ to the horizontal. The rod is in equilibrium. Find the magnitude of the force F .

Diagram: A uniform rod of length L and mass M is pivoted at the left end. A force F is applied to the right end, directed at an angle θ to the horizontal. The rod is in equilibrium. The center of mass is at the midpoint of the rod. The normal force acts at the pivot. The weight acts at the center of mass.



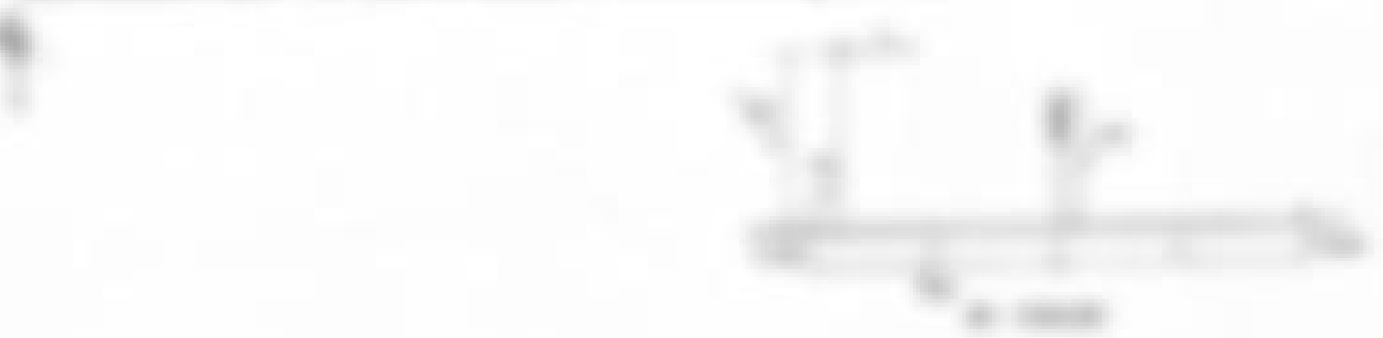
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[illegible]

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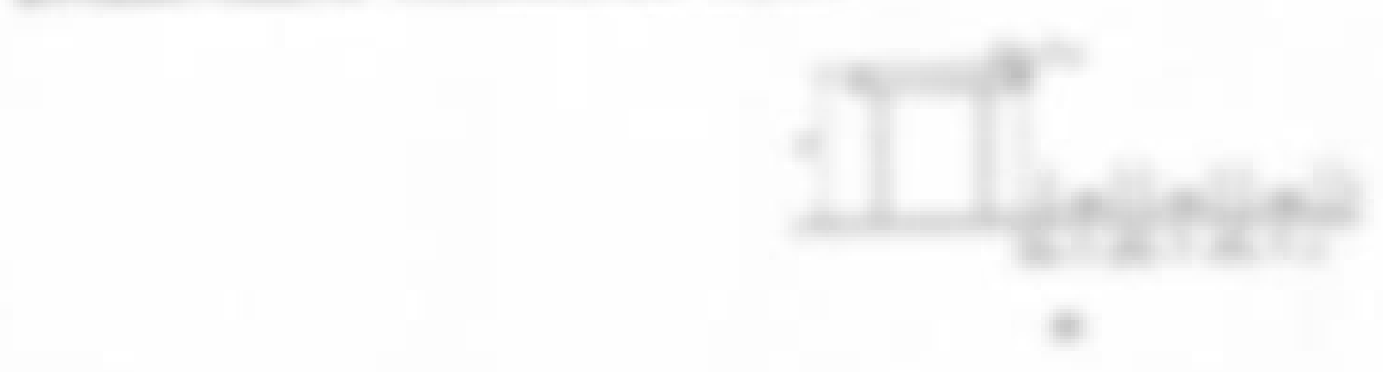
1. The following information is taken from the financial statements of a company for the year ended 31 December 2017:

Revenue 1000000



2. The following information is taken from the financial statements of a company for the year ended 31 December 2017:

- (a) Revenue 1000000
- (b) Cost of sales 600000
- (c) Selling expenses 50000
- (d) Administrative expenses 30000
- (e) Depreciation 20000
- (f) Interest on bank loan 10000
- (g) Profit before tax 100000
- (h) Tax on profit 20000
- (i) Profit after tax 80000



1. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the right at the top edge of the block. The block is on the verge of tipping clockwise about its right edge. Determine the magnitude of the force F in terms of M and h .

ANSWER: $F = \frac{Mh}{2}$

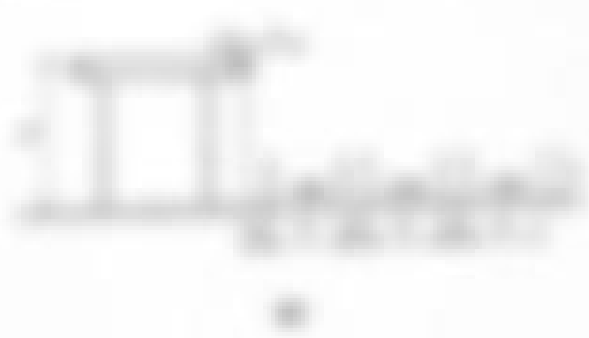


2. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the right at the top edge of the block. The block is on the verge of tipping clockwise about its right edge. Determine the magnitude of the force F in terms of M and h .

ANSWER: $F = \frac{Mh}{2}$

3. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the right at the top edge of the block. The block is on the verge of tipping clockwise about its right edge. Determine the magnitude of the force F in terms of M and h .

ANSWER: $F = \frac{Mh}{2}$



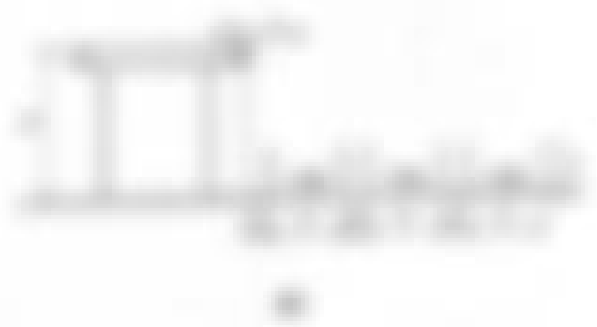
1. **Gegeben:** Ein Balken mit einer Länge von $l = 10\text{ m}$ und einem Gewicht von $G = 10\text{ kN}$. Der Balken ist an einem festen Auflager an der linken Seite und an einem beweglichen Auflager an der rechten Seite. Ein Gewicht $P = 20\text{ kN}$ wirkt senkrecht nach unten auf den Balken in einem Abstand von $a = 3\text{ m}$ von der linken Seite. Berechnen Sie die Auflagerkräfte.



2. **Gegeben:** Ein Balken mit einer Länge von $l = 10\text{ m}$ und einem Gewicht von $G = 10\text{ kN}$. Der Balken ist an einem festen Auflager an der linken Seite und an einem beweglichen Auflager an der rechten Seite. Ein Gewicht $P = 20\text{ kN}$ wirkt senkrecht nach unten auf den Balken in einem Abstand von $a = 3\text{ m}$ von der linken Seite. Berechnen Sie die Auflagerkräfte.

1. **Gegeben:** Ein Balken mit einer Länge von $l = 10\text{ m}$ und einem Gewicht von $G = 10\text{ kN}$. Der Balken ist an einem festen Auflager an der linken Seite und an einem beweglichen Auflager an der rechten Seite. Ein Gewicht $P = 20\text{ kN}$ wirkt senkrecht nach unten auf den Balken in einem Abstand von $a = 3\text{ m}$ von der linken Seite. Berechnen Sie die Auflagerkräfte.

2. **Gegeben:** Ein Balken mit einer Länge von $l = 10\text{ m}$ und einem Gewicht von $G = 10\text{ kN}$. Der Balken ist an einem festen Auflager an der linken Seite und an einem beweglichen Auflager an der rechten Seite. Ein Gewicht $P = 20\text{ kN}$ wirkt senkrecht nach unten auf den Balken in einem Abstand von $a = 3\text{ m}$ von der linken Seite. Berechnen Sie die Auflagerkräfte.



1. Perhatikan gambar berikut ini!

Sebuah benda bergerak lurus berubah beraturan.

- a. $v = 2t$
- b. $v = 4t$
- c. $v = 6t$
- d. $v = 8t$
- e. $v = 10t$



2. Perhatikan gambar berikut ini!

Sebuah benda bergerak lurus berubah beraturan. Grafik kecepatan-waktu (v-t) benda tersebut ditunjukkan di bawah ini.

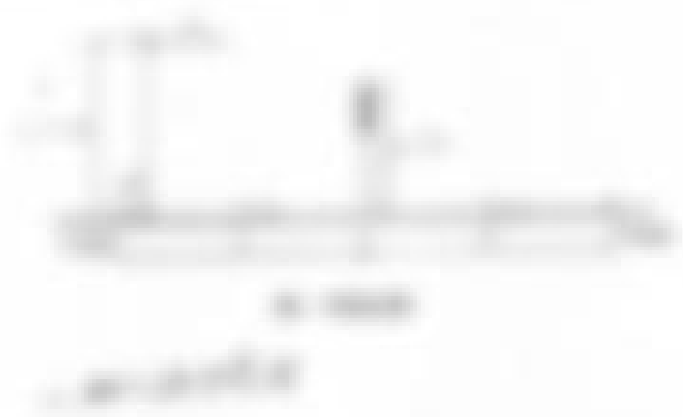
- a. $v = 2t$
- b. $v = 4t$
- c. $v = 6t$
- d. $v = 8t$
- e. $v = 10t$



1. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the x -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in length ΔL of the block.

ANSWER: The change in length ΔL is given by:

$$\Delta L = \frac{\sigma L}{E}$$

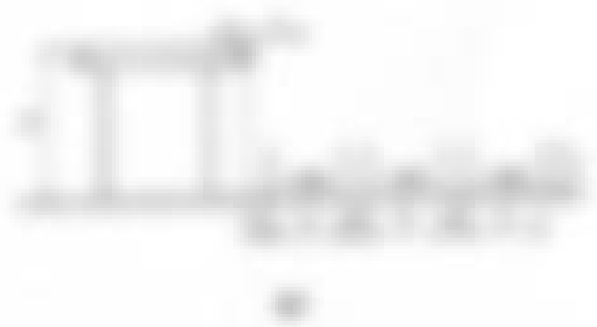


2. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the y -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in width Δb of the block.

ANSWER: The change in width Δb is given by:

QUESTION: A rectangular block of material is subjected to a uniform stress σ in the z -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in height Δh of the block.

$$\Delta h = \frac{\sigma h}{E}$$



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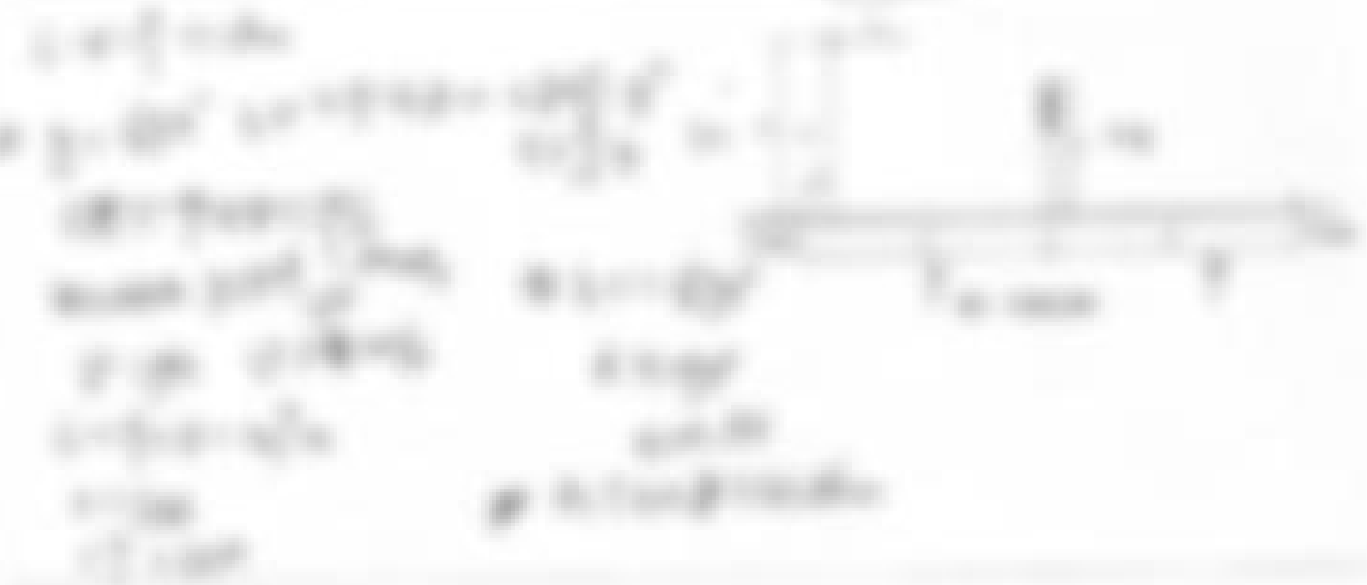
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1. **Abstract:** This paper presents a new approach to the problem of finding the optimal path in a graph. The approach is based on the use of a heuristic function that estimates the cost of the path from the current node to the goal node. The heuristic function is used to select the next node to visit, and the process is repeated until the goal node is reached. The approach is shown to be more efficient than other methods, and is able to find the optimal path in a large number of cases.

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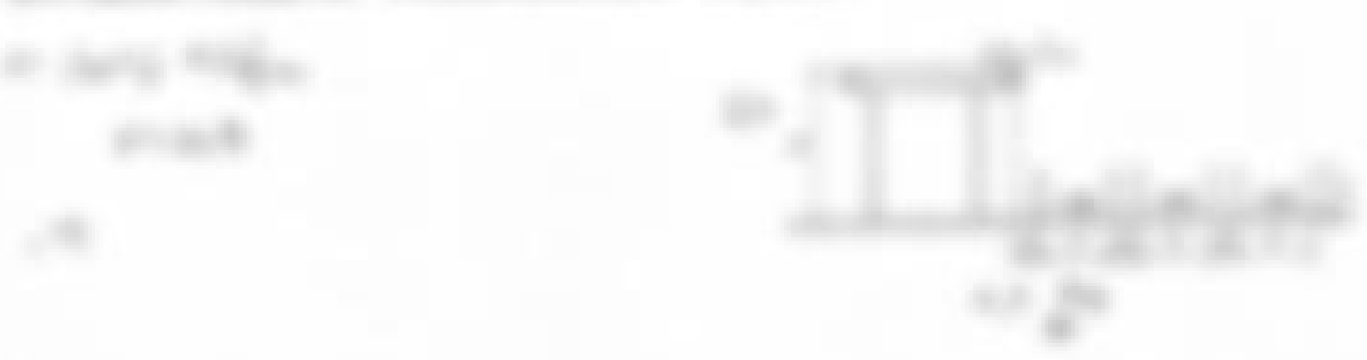
1. **QUESTION:** A horizontal beam of length L is supported at its left end by a pin support and at its right end by a roller support. A uniformly distributed load of intensity w acts downwards over the entire length of the beam. Determine the reaction forces at the supports.

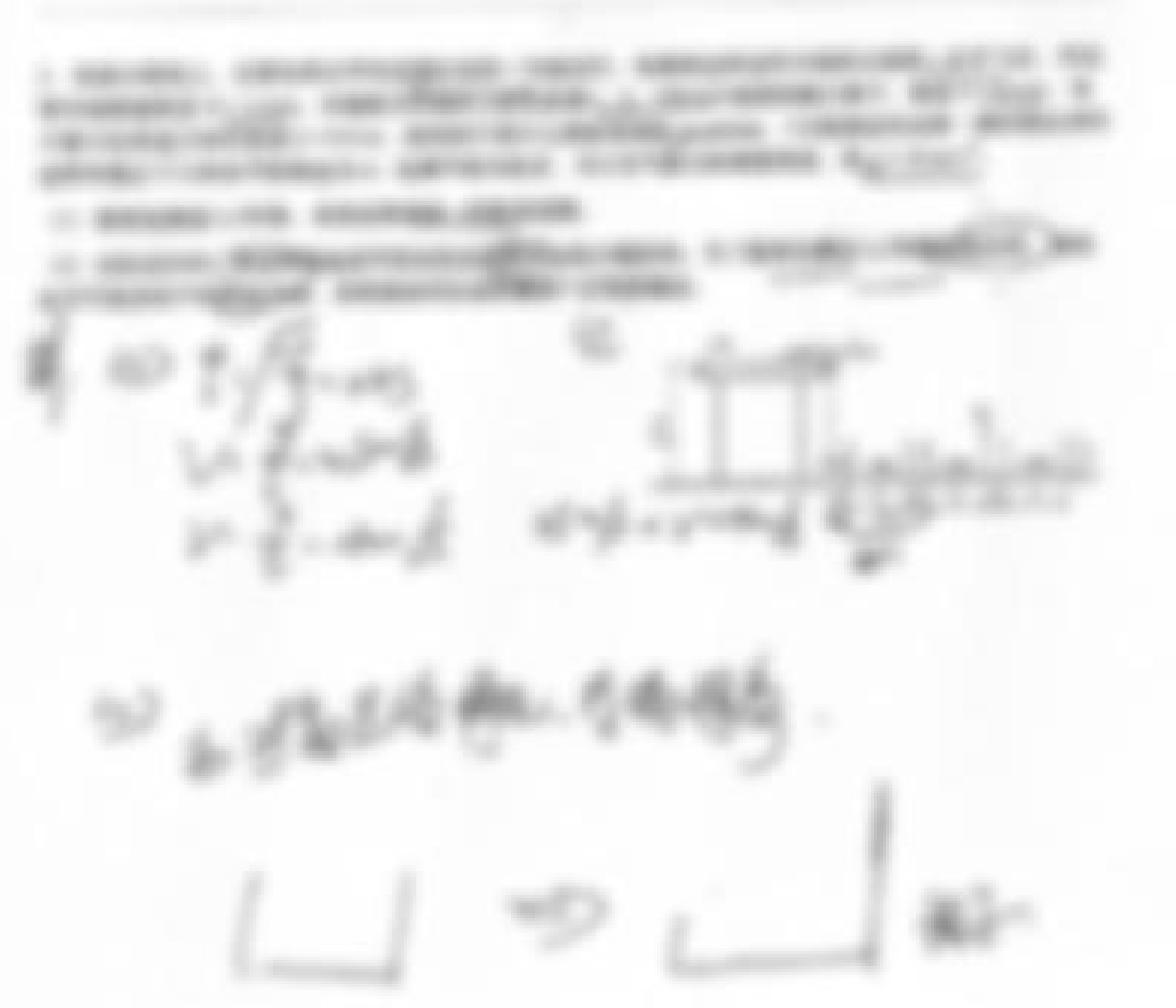
SOLUTION: Draw a free-body diagram of the beam.



2. **QUESTION:** A horizontal beam of length L is supported at its left end by a pin support and at its right end by a roller support. A point load of magnitude P acts downwards at a distance a from the left end. Determine the reaction forces at the supports.

SOLUTION: Draw a free-body diagram of the beam.





1. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.

2. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.



3. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.

4. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.



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1. The first step is to identify the problem. This involves understanding the current situation and what needs to be changed.

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1. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,582,000. The population of the United States in 1950 was 150,697,000. The population of the United States in 1940 was 132,165,000. The population of the United States in 1930 was 122,766,000. The population of the United States in 1920 was 105,950,000. The population of the United States in 1910 was 92,228,000. The population of the United States in 1900 was 76,212,000. The population of the United States in 1890 was 62,948,000. The population of the United States in 1880 was 50,189,000. The population of the United States in 1870 was 38,628,000. The population of the United States in 1860 was 31,443,000. The population of the United States in 1850 was 23,802,000. The population of the United States in 1840 was 17,963,000. The population of the United States in 1830 was 12,866,000. The population of the United States in 1820 was 9,637,000. The population of the United States in 1810 was 7,267,000. The population of the United States in 1800 was 5,308,000. The population of the United States in 1790 was 3,929,000. The population of the United States in 1780 was 2,939,000. The population of the United States in 1770 was 2,147,000. The population of the United States in 1760 was 1,584,000. The population of the United States in 1750 was 1,193,000. The population of the United States in 1740 was 884,000. The population of the United States in 1730 was 659,000. The population of the United States in 1720 was 496,000. The population of the United States in 1710 was 375,000. The population of the United States in 1700 was 281,000. The population of the United States in 1690 was 214,000. The population of the United States in 1680 was 163,000. The population of the United States in 1670 was 125,000. The population of the United States in 1660 was 96,000. The population of the United States in 1650 was 73,000. The population of the United States in 1640 was 56,000. The population of the United States in 1630 was 43,000. The population of the United States in 1620 was 33,000. The population of the United States in 1610 was 25,000. The population of the United States in 1600 was 19,000. The population of the United States in 1590 was 14,000. The population of the United States in 1580 was 11,000. The population of the United States in 1570 was 8,000. The population of the United States in 1560 was 6,000. The population of the United States in 1550 was 4,000. The population of the United States in 1540 was 3,000. The population of the United States in 1530 was 2,000. The population of the United States in 1520 was 1,000. The population of the United States in 1510 was 500. The population of the United States in 1500 was 250. The population of the United States in 1490 was 125. The population of the United States in 1480 was 62.5. The population of the United States in 1470 was 31.25. The population of the United States in 1460 was 15.625. The population of the United States in 1450 was 7.8125. The population of the United States in 1440 was 3.90625. The population of the United States in 1430 was 1.953125. The population of the United States in 1420 was 0.9765625. The population of the United States in 1410 was 0.48828125. The population of the United States in 1400 was 0.244140625. The population of the United States in 1390 was 0.1220703125. The population of the United States in 1380 was 0.06103515625. The population of the United States in 1370 was 0.030517578125. The population of the United States in 1360 was 0.0152587890625. The population of the United States in 1350 was 0.00762939453125. The population of the United States in 1340 was 0.003814697265625. The population of the United States in 1330 was 0.0019073486328125. The population of the United States in 1320 was 0.00095367431640625. 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The population of the United States in 410 was 3.85185988877447016420774161815643310546875e-31. The population of the United States in 400 was 1.925929944387235082103870809078216552734375e-31. The population of the United States in 390 was 9.629649721936175410519354045391082763671875e-32. The population of the United States in 380 was 4.8148248609680877052596770226955413818359375e-32. The population of the United States in 370 was 2.40741243048404385262983851134777069091796875e-32. The population of the United States in 360 was 1.203706215242021926314919255673885345458984375e-32. The population of the United States in 350 was 6.018531076210109631574596278369426727294921875e-33. The population of the United States in 340 was 3.0092655381050548157872981391847133636474609375e-33. The population of the United States in 330 was 1.50463276905252740789364906959235668182373046875e-33. The population of the United States in 320 was 7.52316384526263703946824534796178340911865234375e-34. 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The population of the United States in 220 was 7.346839692639293983855708347618929110467433929443359375e-37. The population of the United States in 210 was 3.6734198463196469919278541738094645552337169647216796875e-37. The population of the United States in 200 was 1.83670992315982349596392708690473227761685848236083984375e-37. The population of the United States in 190 was 9.18354961579911747981963543452366138808429241180419921875e-38. The population of the United States in 180 was 4.591774807899558739909817717261830694042146205902099609375e-38. The population of the United States in 170 was 2.2958874039497793699549088586309153470210731029510498046875e-38. The population of the United States in 160 was 1.14794370197488968497745442931545767351053655147552490234375e-38. The population of the United States in 150 was 5.73971850987444842488727214657728836755268275737762451171875e-39. The population of the United States in 140 was 2.869859254937224212443636073288644183776341378688812255859375e-39. The population of the United States in 130 was 1.4349296274686121062218180366443220918881706893444061279296875e-39. The population of the United States in 120 was 7.1746481373430605311090901832216104594408534467220306396484375e-40. The population of the United States in 110 was 3.58732406867153026555454509161080522972042672336101531982421875e-40. The population of the United States in 100 was 1.793662034335765132777272545805402614860213361680507659912109375e-40. The population of the United States in 90 was 8.968310171678825663886362729027013074301066808402538299560546875e-41. The population of the United States in 80 was 4.4841550858394128319431813645135065371505334042012691497802734375e-41. The population of the United States in 70 was 2.24207754291970641597159068225675326857526670210063457489013671875e-41. The population of the United States in 60 was 1.121038771459853207985795341128376634287633351050317287445068359375e-41. The population of the United States in 50 was 5.605193857299266039928976705641883171438166755251586437225341796875e-42. The population of the United States in 40 was 2.8025969286496330199644883528209415857190833776257932186126708984375e-42. The population of the United States in 30 was 1.40129846432481650998224417641047079285954168881289660930633544921875e-42. The population of the United States in 20 was 7.00649232162408254991122088205235396429770844406448304653167724609375e-43. The population of the United States in 10 was 3.503246160812041274955610441026176982148854222032241523265838623046875e-43. The population of the United States in 0 was 1.7516230804060206374778052205130884910744271110161207616329193115234375e-43.

1. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,582,000. The population of the United States in 1950 was 150,697,000. The population of the United States in 1940 was 132,165,000. The population of the United States in 1930 was 122,766,000. The population of the United States in 1920 was 105,950,000. The population of the United States in 1910 was 92,228,000. The population of the United States in 1900 was 76,212,000. The population of the United States in 1890 was 62,948,000. The population of the United States in 1880 was 50,189,000. The population of the United States in 1870 was 38,628,000. The population of the United States in 1860 was 31,443,000. The population of the United States in 1850 was 23,802,000. The population of the United States in 1840 was 17,963,000. The population of the United States in 1830 was 12,866,000. The population of the United States in 1820 was 9,637,000. The population of the United States in 1810 was 7,267,000. The population of the United States in 1800 was 5,308,000. The population of the United States in 1790 was 3,929,000. The population of the United States in 1780 was 2,939,000. The population of the United States in 1770 was 2,147,000. The population of the United States in 1760 was 1,584,000. The population of the United States in 1750 was 1,193,000. The population of the United States in 1740 was 884,000. The population of the United States in 1730 was 659,000. The population of the United States in 1720 was 496,000. The population of the United States in 1710 was 375,000. The population of the United States in 1700 was 281,000. The population of the United States in 1690 was 214,000. The population of the United States in 1680 was 163,000. The population of the United States in 1670 was 125,000. The population of the United States in 1660 was 96,000. The population of the United States in 1650 was 73,000. The population of the United States in 1640 was 56,000. The population of the United States in 1630 was 43,000. The population of the United States in 1620 was 33,000. The population of the United States in 1610 was 25,000. The population of the United States in 1600 was 19,000. The population of the United States in 1590 was 14,000. The population of the United States in 1580 was 11,000. The population of the United States in 1570 was 8,000. The population of the United States in 1560 was 6,000. The population of the United States in 1550 was 4,000. The population of the United States in 1540 was 3,000. The population of the United States in 1530 was 2,000. The population of the United States in 1520 was 1,000. The population of the United States in 1510 was 500. The population of the United States in 1500 was 250. The population of the United States in 1490 was 125. The population of the United States in 1480 was 62.5. The population of the United States in 1470 was 31.25. The population of the United States in 1460 was 15.625. The population of the United States in 1450 was 7.8125. The population of the United States in 1440 was 3.90625. The population of the United States in 1430 was 1.953125. The population of the United States in 1420 was 0.9765625. The population of the United States in 1410 was 0.48828125. The population of the United States in 1400 was 0.244140625. The population of the United States in 1390 was 0.1220703125. The population of the United States in 1380 was 0.06103515625. The population of the United States in 1370 was 0.030517578125. The population of the United States in 1360 was 0.0152587890625. The population of the United States in 1350 was 0.00762939453125. The population of the United States in 1340 was 0.003814697265625. The population of the United States in 1330 was 0.0019073486328125. The population of the United States in 1320 was 0.00095367431640625. The population of the United States in 1310 was 0.000476837158203125. The population of the United States in 1300 was 0.0002384185791015625. The population of the United States in 1290 was 0.00011920928955078125. The population of the United States in 1280 was 5.960464477539062e-05. The population of the United States in 1270 was 2.980232238769531e-05. The population of the United States in 1260 was 1.490116119384766e-05. The population of the United States in 1250 was 7.450580596923828e-06. The population of the United States in 1240 was 3.725290298461914e-06. The population of the United States in 1230 was 1.862645149230957e-06. The population of the United States in 1220 was 9.313225746154785e-07. The population of the United States in 1210 was 4.656612873077392e-07. The population of the United States in 1200 was 2.328306436538696e-07. The population of the United States in 1190 was 1.164153218269348e-07. The population of the United States in 1180 was 5.82076609134674e-08. The population of the United States in 1170 was 2.91038304567337e-08. The population of the United States in 1160 was 1.455191522836685e-08. The population of the United States in 1150 was 7.275957614183425e-09. The population of the United States in 1140 was 3.637978807091712e-09. The population of the United States in 1130 was 1.818989403545856e-09. The population of the United States in 1120 was 9.09494701772928e-10. The population of the United States in 1110 was 4.54747350886464e-10. The population of the United States in 1100 was 2.27373675443232e-10. The population of the United States in 1090 was 1.13686837721616e-10. The population of the United States in 1080 was 5.6843418860808e-11. The population of the United States in 1070 was 2.8421709430404e-11. The population of the United States in 1060 was 1.4210854715202e-11. The population of the United States in 1050 was 7.105427357601e-12. The population of the United States in 1040 was 3.5527136788005e-12. The population of the United States in 1030 was 1.77635683940025e-12. The population of the United States in 1020 was 8.88178419700125e-13. The population of the United States in 1010 was 4.440892098500625e-13. The population of the United States in 1000 was 2.2204460492503125e-13. The population of the United States in 990 was 1.110223024625156e-13. The population of the United States in 980 was 5.55111512312578e-14. The population of the United States in 970 was 2.77555756156289e-14. The population of the United States in 960 was 1.387778780781445e-14. The population of the United States in 950 was 6.938893903907225e-15. The population of the United States in 940 was 3.4694469519536125e-15. The population of the United States in 930 was 1.734723475976806e-15. The population of the United States in 920 was 8.67361737988403e-16. The population of the United States in 910 was 4.336808689942015e-16. The population of the United States in 900 was 2.168404344971007e-16. The population of the United States in 890 was 1.084202172485504e-16. The population of the United States in 880 was 5.42101086242752e-17. The population of the United States in 870 was 2.71050543121376e-17. The population of the United States in 860 was 1.35525271560688e-17. The population of the United States in 850 was 6.7762635780344e-18. The population of the United States in 840 was 3.3881317890172e-18. The population of the United States in 830 was 1.6940658945086e-18. The population of the United States in 820 was 8.470329472543e-19. The population of the United States in 810 was 4.2351647362715e-19. The population of the United States in 800 was 2.11758236813575e-19. The population of the United States in 790 was 1.058791184067875e-19. The population of the United States in 780 was 5.293955920339375e-20. The population of the United States in 770 was 2.6469779601696875e-20. The population of the United States in 760 was 1.32348898008484375e-20. The population of the United States in 750 was 6.61744490042421875e-21. The population of the United States in 740 was 3.308722450212109375e-21. The population of the United States in 730 was 1.6543612251060546875e-21. The population of the United States in 720 was 8.2718061255302734375e-22. The population of the United States in 710 was 4.13590306276513671875e-22. The population of the United States in 700 was 2.067951531382568359375e-22. The population of the United States in 690 was 1.0339757656912841796875e-22. The population of the United States in 680 was 5.1698788284564208984375e-23. The population of the United States in 670 was 2.58493941422821044921875e-23. The population of the United States in 660 was 1.292469707114105224609375e-23. The population of the United States in 650 was 6.462348535570526123046875

[illegible]

1. **Abstract:** This paper presents a new approach to the problem of finding the optimal path in a graph. The approach is based on the use of a heuristic function that estimates the cost of the path from the current node to the goal node. The heuristic function is used to select the next node to visit, and the process is repeated until the goal node is reached. The approach is shown to be more efficient than other methods, and is able to find the optimal path in a large number of cases.

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1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .

4. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .



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2. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose. It is not to be used for any other purpose.

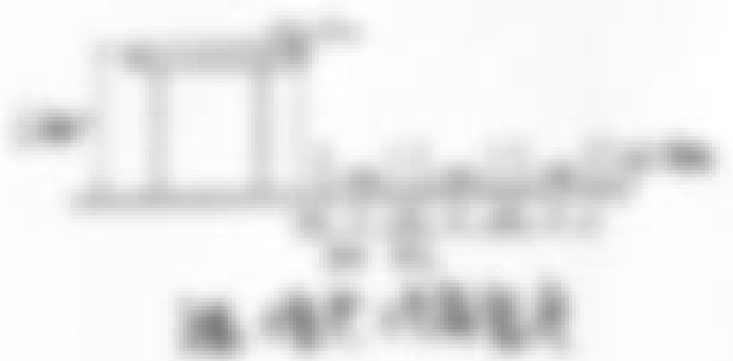
3. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose. It is not to be used for any other purpose.



4. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose. It is not to be used for any other purpose.

5. The following information is for your information only. It is not to be used for any other purpose. It is not to be used for any other purpose. It is not to be used for any other purpose.

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4 3

1. A particle of mass m moves in a circular path of radius r with a constant speed v . The centripetal force acting on the particle is F_c . The work done by the centripetal force in one complete revolution is W .

$t = 4.95$

$g = 10 \text{ m/s}^2$

$v = 20 \text{ m/s}$

(wt. 30%)



$t = 0.1 \text{ s}$

$v = 10 \text{ m/s}$

2. A particle of mass m moves in a circular path of radius r with a constant speed v . The centripetal force acting on the particle is F_c . The work done by the centripetal force in one complete revolution is W .

- a. Calculate the centripetal force.
- b. Calculate the work done by the centripetal force in one complete revolution.

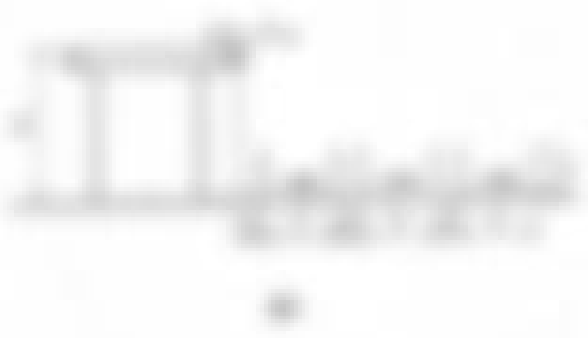


1. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.



2. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.

3. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v .

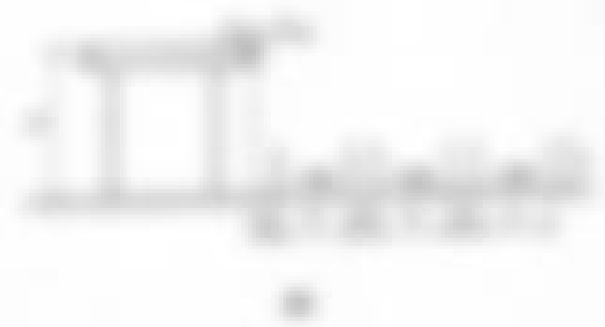
2. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v .



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v .

4. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v .

5. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v .



1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

Answer: $F = \frac{mgh}{2}$

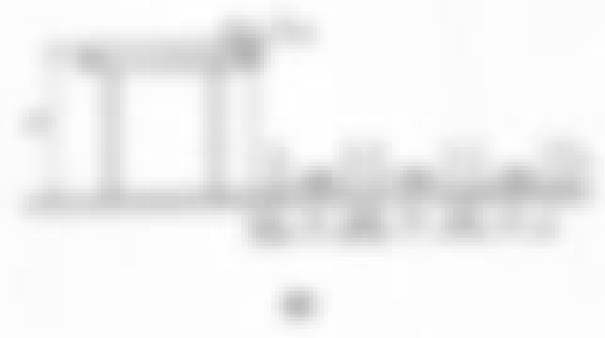


2. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

Answer: $F = \frac{mgh}{2}$

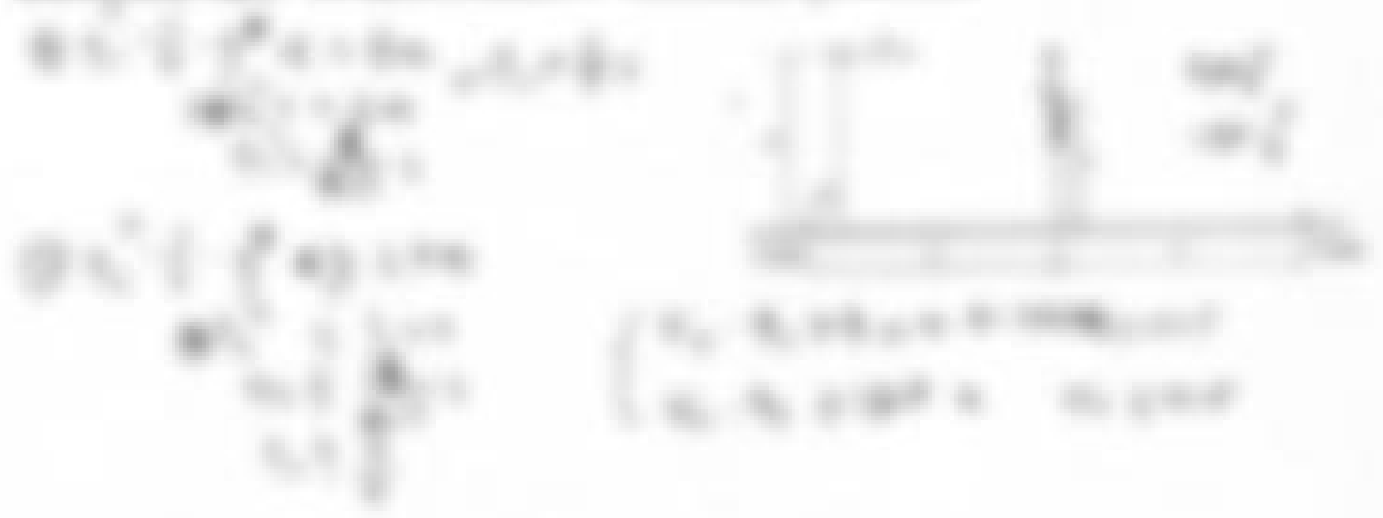
3. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

Answer: $F = \frac{mgh}{2}$



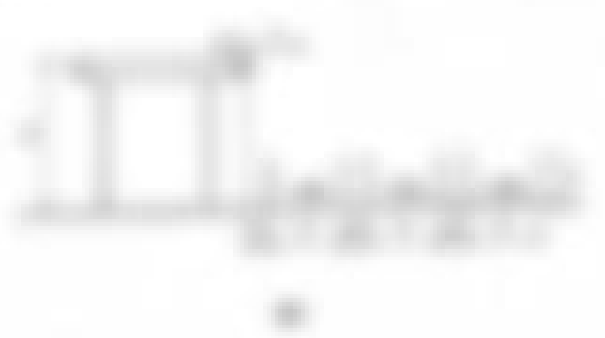
1. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following table shows the distribution of the population by age group and sex.

Population by Age Group and Sex



2. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following table shows the distribution of the population by age group and sex.

- a. The following table shows the distribution of the population by age group and sex.
- b. The following table shows the distribution of the population by age group and sex.



1. The following information is for your information only. It is not intended to be used as a basis for any action.

2. The following information is for your information only. It is not intended to be used as a basis for any action.



3. The following information is for your information only. It is not intended to be used as a basis for any action.

4. The following information is for your information only. It is not intended to be used as a basis for any action.



5. The following information is for your information only. It is not intended to be used as a basis for any action.

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1. A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. The distance between the supports is L .

2. Determine the reaction forces at the supports.

$b = 4\text{ m}$ $h = 2\text{ m}$
 $q = 10\text{ kN/m}$ $L = 6\text{ m}$
 $R_1 = ?$ $R_2 = ?$
 $R_3 = ?$ $R_4 = ?$



3. A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. The distance between the supports is L .

4. Determine the reaction forces at the supports.

$b = 4\text{ m}$ $h = 2\text{ m}$



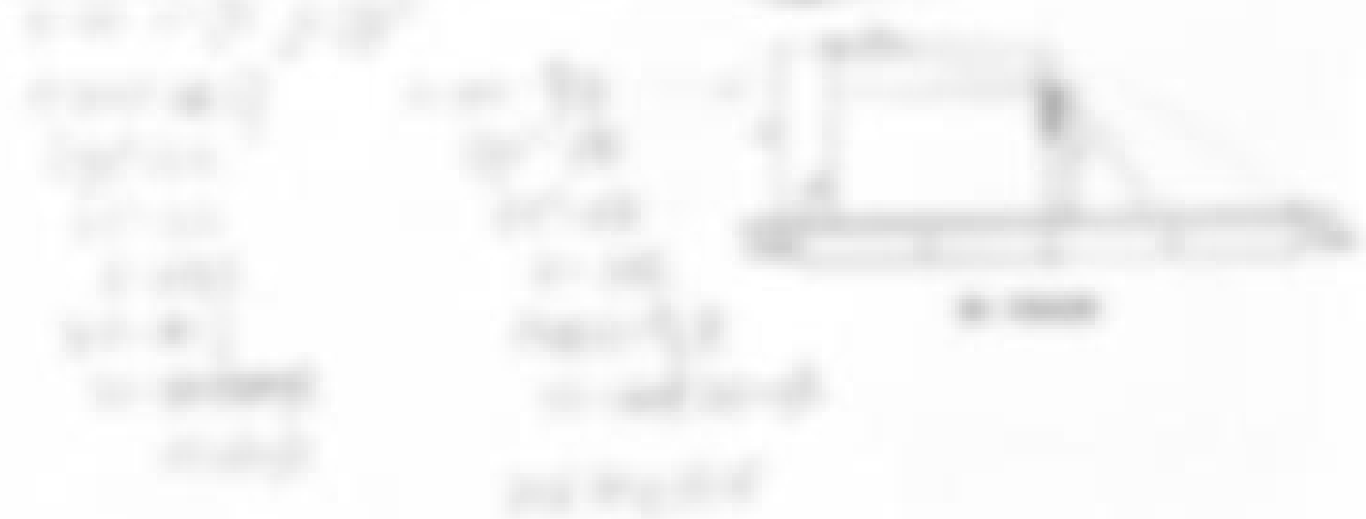
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1. **Abstract:** This paper presents a new method for the automatic detection of the onset of a seizure. The method is based on the analysis of the non-linear properties of the EEG signal. The results show that the proposed method is able to detect the onset of a seizure with a high accuracy and a low false alarm rate.

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1. **PROBLEM STATEMENT:** A rectangular block of material is subjected to a uniform shear stress τ applied parallel to its top and bottom surfaces. The block is fixed to a rigid base.

ASSUMPTIONS: The material is homogeneous and isotropic. The deformation is small.



2. **ANALYSIS:** The shear stress τ is constant throughout the block. The shear strain γ is constant throughout the block. The shear strain γ is defined as the angle of deformation. The shear strain γ is related to the shear stress τ by the shear modulus G of the material.

CONCLUSIONS: The shear stress τ is constant throughout the block. The shear strain γ is constant throughout the block. The shear strain γ is related to the shear stress τ by the shear modulus G of the material.





1. The following information is given for the first three months of the year 2000. The company has a fixed production overhead of \$100,000 per month.

Required: Calculate the fixed production overhead variance for the first three months of the year 2000.

Information
Actual production 10,000 units
Standard production 12,000 units
Standard cost per unit \$10



2. The following information is given for the first three months of the year 2000. The company has a fixed production overhead of \$100,000 per month.

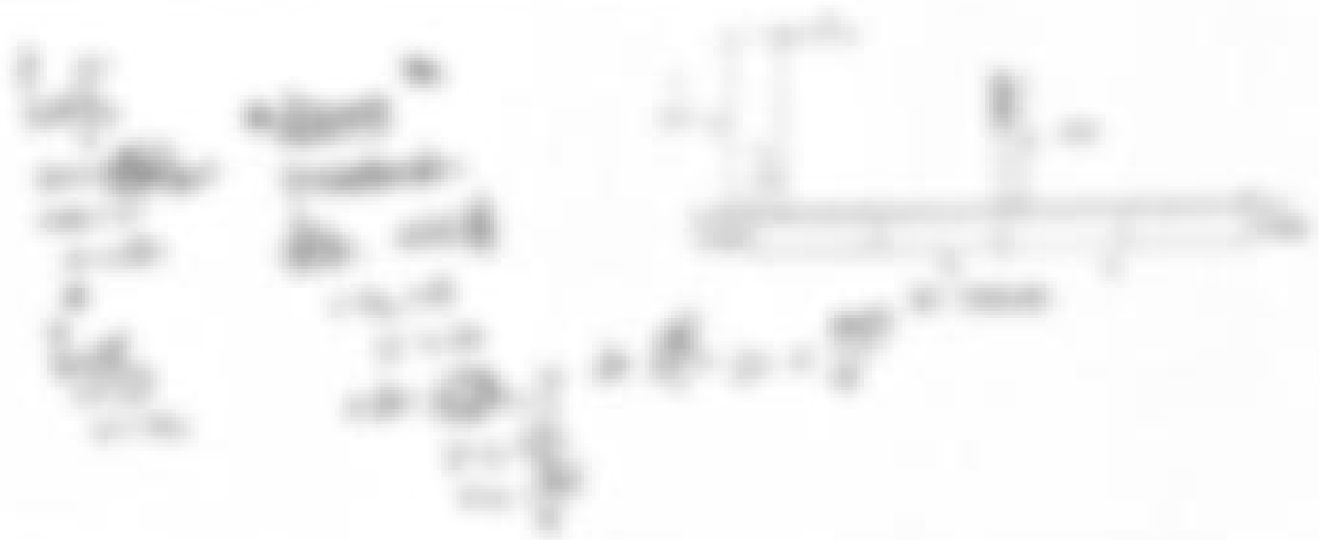
Required: Calculate the fixed production overhead variance for the first three months of the year 2000.

3. The following information is given for the first three months of the year 2000. The company has a fixed production overhead of \$100,000 per month.



1. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 179,329,453.

Source: U.S. Census Bureau, 1960.



2. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 179,329,453. The population of the United States in 1950 was 152,266,000. The population of the United States in 1940 was 132,165,000. The population of the United States in 1930 was 122,766,000. The population of the United States in 1920 was 105,710,000. The population of the United States in 1910 was 92,228,000. The population of the United States in 1900 was 76,212,000.

Source: U.S. Census Bureau, 1960.



Population in millions
1900 76.2
1910 92.2
1920 105.7
1930 122.8
1940 132.2
1950 152.3
1960 179.3

1. **QUESTION** (10 marks) (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The horizontal distance between O and A is R . The time taken for the particle to reach A is T . The maximum height reached by the particle is H . The horizontal range R is given by $R = \frac{u^2 \sin 2\theta}{g}$. The time of flight T is given by $T = \frac{2u \sin \theta}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.

ANSWER (10 marks) (10 marks)

1. **QUESTION** (10 marks) (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The horizontal distance between O and A is R . The time taken for the particle to reach A is T . The maximum height reached by the particle is H . The horizontal range R is given by $R = \frac{u^2 \sin 2\theta}{g}$. The time of flight T is given by $T = \frac{2u \sin \theta}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.



2. **QUESTION** (10 marks) (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The horizontal distance between O and A is R . The time taken for the particle to reach A is T . The maximum height reached by the particle is H . The horizontal range R is given by $R = \frac{u^2 \sin 2\theta}{g}$. The time of flight T is given by $T = \frac{2u \sin \theta}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.

ANSWER (10 marks) (10 marks)
2. **QUESTION** (10 marks) (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The horizontal distance between O and A is R . The time taken for the particle to reach A is T . The maximum height reached by the particle is H . The horizontal range R is given by $R = \frac{u^2 \sin 2\theta}{g}$. The time of flight T is given by $T = \frac{2u \sin \theta}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.

3. **QUESTION** (10 marks) (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial velocity u at an angle θ to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The horizontal distance between O and A is R . The time taken for the particle to reach A is T . The maximum height reached by the particle is H . The horizontal range R is given by $R = \frac{u^2 \sin 2\theta}{g}$. The time of flight T is given by $T = \frac{2u \sin \theta}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \theta}{2g}$.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v and the strain in the horizontal direction is denoted by ϵ_h . The Poisson's ratio of the material is denoted by ν . The relationship between ϵ_v and ϵ_h is given by:

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v and the strain in the horizontal direction is denoted by ϵ_h . The Poisson's ratio of the material is denoted by ν . The relationship between ϵ_v and ϵ_h is given by:

- (a) $\epsilon_h = \nu \epsilon_v$
- (b) $\epsilon_h = -\nu \epsilon_v$
- (c) $\epsilon_h = \epsilon_v$
- (d) $\epsilon_h = -\epsilon_v$
- (e) $\epsilon_h = \nu \epsilon_v + \epsilon_v$
- (f) $\epsilon_h = -\nu \epsilon_v - \epsilon_v$
- (g) $\epsilon_h = \nu \epsilon_v - \epsilon_v$
- (h) $\epsilon_h = -\nu \epsilon_v + \epsilon_v$



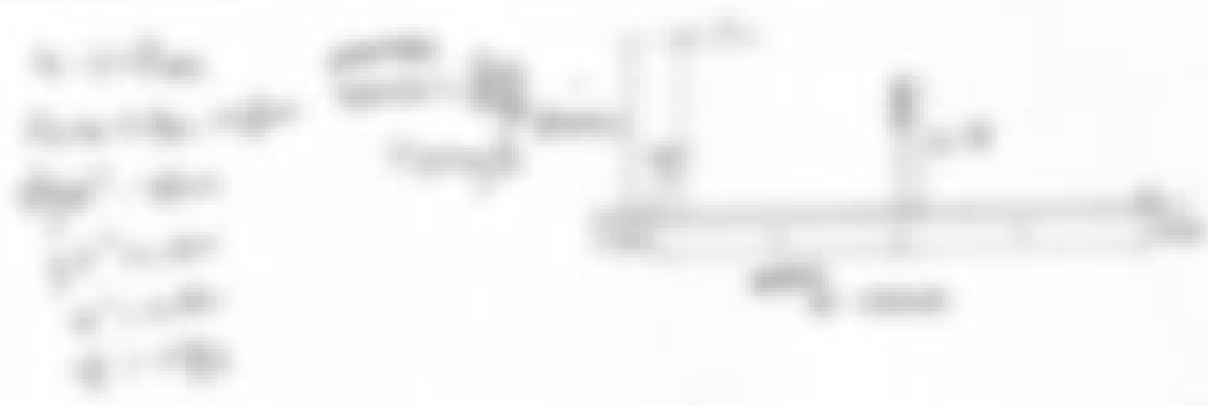
3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The strain in the vertical direction is denoted by ϵ_v and the strain in the horizontal direction is denoted by ϵ_h . The Poisson's ratio of the material is denoted by ν . The relationship between ϵ_v and ϵ_h is given by:

- (a) $\epsilon_h = \nu \epsilon_v$
- (b) $\epsilon_h = -\nu \epsilon_v$
- (c) $\epsilon_h = \epsilon_v$
- (d) $\epsilon_h = -\epsilon_v$
- (e) $\epsilon_h = \nu \epsilon_v + \epsilon_v$
- (f) $\epsilon_h = -\nu \epsilon_v - \epsilon_v$
- (g) $\epsilon_h = \nu \epsilon_v - \epsilon_v$
- (h) $\epsilon_h = -\nu \epsilon_v + \epsilon_v$



1. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three groups: A, B, and C. The population of group A was 100,000, the population of group B was 200,000, and the population of group C was 300,000.

2. The following information was obtained from a survey of the population of a certain country in 1990.



3. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three groups: A, B, and C. The population of group A was 100,000, the population of group B was 200,000, and the population of group C was 300,000.

4. The following information was obtained from a survey of the population of a certain country in 1990.

5. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three groups: A, B, and C. The population of group A was 100,000, the population of group B was 200,000, and the population of group C was 300,000.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .

2. The displacement δ is given by the following expression:

- (a) $\delta = \frac{\sigma L}{E}$
- (b) $\delta = \frac{\sigma L}{2E}$
- (c) $\delta = \frac{\sigma L}{3E}$
- (d) $\delta = \frac{\sigma L}{4E}$



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The top part is fixed to a rigid wall. The bottom part is free to move. The displacement of the bottom part is denoted by δ .

- (a) $\delta = \frac{\sigma L}{E}$
- (b) $\delta = \frac{\sigma L}{2E}$
- (c) $\delta = \frac{\sigma L}{3E}$
- (d) $\delta = \frac{\sigma L}{4E}$



1. A particle of mass m moves in a circular path of radius r with a constant speed v . The centripetal force acting on the particle is F_c . The work done by the centripetal force in one complete revolution is W . Find W .

Solution: Centripetal force is always perpendicular to the displacement. Hence, the work done is zero.

$$\begin{aligned} F_c &= \frac{mv^2}{r} \\ W &= F_c \cdot r \cdot \cos 90^\circ \\ W &= \frac{mv^2}{r} \cdot r \cdot 0 \\ W &= 0 \end{aligned}$$

$$\begin{aligned} F_c &= \frac{mv^2}{r} \\ W &= F_c \cdot r \cdot \cos 90^\circ \\ W &= \frac{mv^2}{r} \cdot r \cdot 0 \\ W &= 0 \end{aligned}$$



Work done = 0

2. A particle of mass m moves in a circular path of radius r with a constant speed v . The centripetal force acting on the particle is F_c . The work done by the centripetal force in one complete revolution is W . Find W .

Solution: Centripetal force is always perpendicular to the displacement. Hence, the work done is zero.

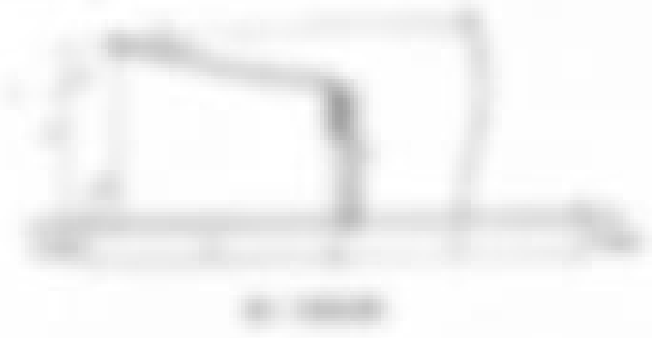
3. A particle of mass m moves in a circular path of radius r with a constant speed v . The centripetal force acting on the particle is F_c . The work done by the centripetal force in one complete revolution is W . Find W .



•

1. The following information was obtained from a survey of the population of a certain country in 1990. The population was 100,000. The number of people in each age group is given in the table below.

Age Group Number of People



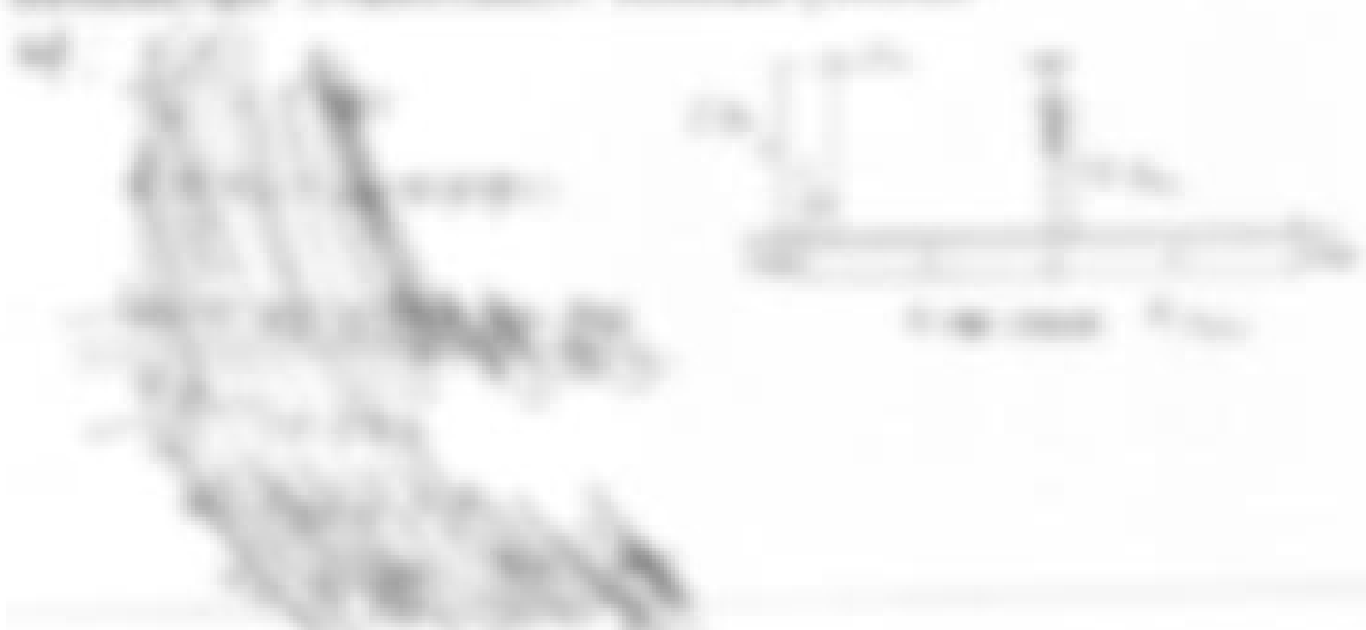
2. The following information was obtained from a survey of the population of a certain country in 1990. The population was 100,000. The number of people in each age group is given in the table below.

Age Group Number of People



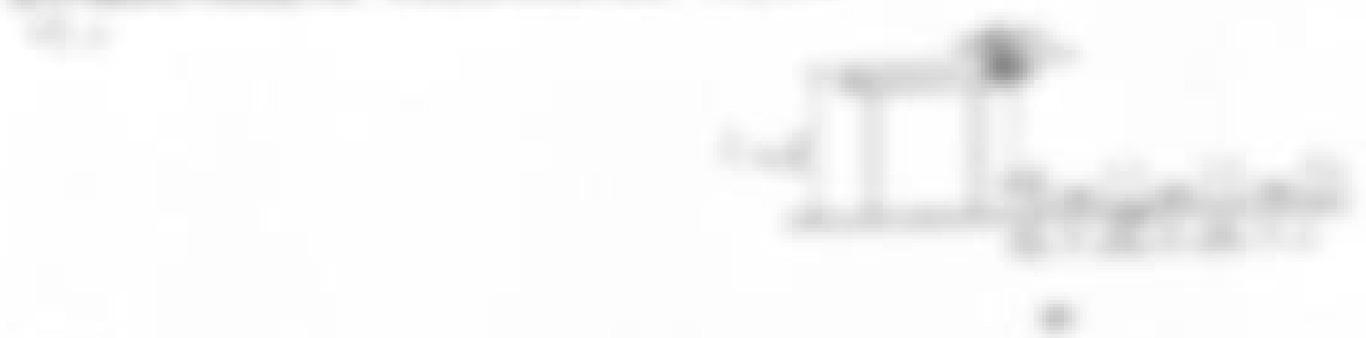
1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ_x in the x -direction and a uniform stress σ_y in the y -direction. The block is also subjected to a uniform shear stress τ_{xy} in the xy -plane. The block is shown in the figure with the stresses indicated by arrows.

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ_x in the x -direction and a uniform stress σ_y in the y -direction. The block is also subjected to a uniform shear stress τ_{xy} in the xy -plane. The block is shown in the figure with the stresses indicated by arrows.



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ_x in the x -direction and a uniform stress σ_y in the y -direction. The block is also subjected to a uniform shear stress τ_{xy} in the xy -plane. The block is shown in the figure with the stresses indicated by arrows.

- (a) Determine the principal stresses σ_1 and σ_2 .
- (b) Determine the maximum shear stress τ_{max} .
- (c) Determine the average normal stress σ_{avg} .



4. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ_x in the x -direction and a uniform stress σ_y in the y -direction. The block is also subjected to a uniform shear stress τ_{xy} in the xy -plane. The block is shown in the figure with the stresses indicated by arrows.

1. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reaction forces at the supports.

2. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.



3. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.

4. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.

5. A horizontal beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the midpoint of the beam. Determine the reaction forces at the supports.



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1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific requirements of the task.

1. **General Information**
2. **Project Description**
3. **Objectives and Scope**
4. **Methodology**
5. **Timeline and Milestones**
6. **Budget and Resources**
7. **Risks and Mitigation**
8. **Conclusion**

1. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.

2. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.

3. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.



4. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.

5. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.

6. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.

7. The following information is for the purpose of providing a general overview of the system. It is not intended to be a detailed description of the system or its components.



1. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.

2. The following information was obtained from a survey of the population of the United States in 1960.

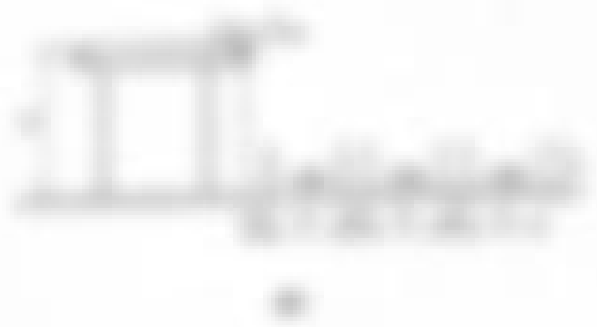
Age	Male	Female
0-4	10,000,000	10,000,000
5-9	9,000,000	9,000,000
10-14	8,000,000	8,000,000
15-19	7,000,000	7,000,000
20-24	6,000,000	6,000,000
25-29	5,000,000	5,000,000
30-34	4,000,000	4,000,000
35-39	3,000,000	3,000,000
40-44	2,000,000	2,000,000
45-49	1,000,000	1,000,000
50-54	1,000,000	1,000,000
55-59	1,000,000	1,000,000
60-64	1,000,000	1,000,000
65-69	1,000,000	1,000,000
70-74	1,000,000	1,000,000
75-79	1,000,000	1,000,000
80-84	1,000,000	1,000,000
85-89	1,000,000	1,000,000
90-94	1,000,000	1,000,000
95-99	1,000,000	1,000,000
100+	1,000,000	1,000,000



3. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.

4. The following information was obtained from a survey of the population of the United States in 1960.

5. The following information was obtained from a survey of the population of the United States in 1960. The population of the United States in 1960 was 192,500,000.



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1. **Abstract:** This paper presents a new method for the automatic detection of the onset of a seizure. The method is based on the analysis of the EEG signal and the detection of the onset of a seizure is achieved by the analysis of the power spectrum of the EEG signal. The method is applied to the analysis of the EEG signal recorded from a patient with a seizure. The results show that the method is able to detect the onset of a seizure with a high accuracy.

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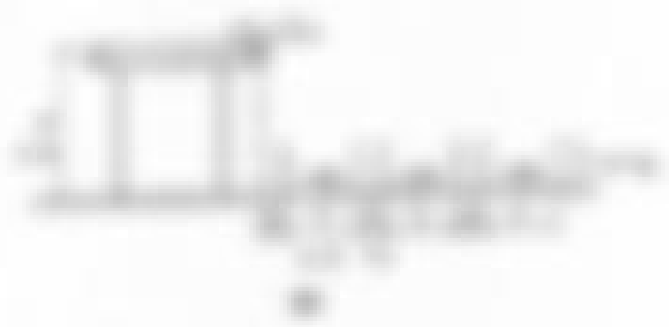
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1. The following information was obtained from a survey of the population of a certain city. The population is divided into four groups, A, B, C, and D, and the following information was obtained:

Group	Population	Income (in thousands of dollars)
A	100	10
B	200	20
C	300	30
D	400	40

2. The following information was obtained from a survey of the population of a certain city. The population is divided into four groups, A, B, C, and D, and the following information was obtained:

- a. The population of group A is 100.
- b. The population of group B is 200.
- c. The population of group C is 300.
- d. The population of group D is 400.



1. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three age groups: 0-14, 15-64, and 65 and over. The population of each age group is given in the table below.



2. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three age groups: 0-14, 15-64, and 65 and over. The population of each age group is given in the table below.

3. The following information was obtained from a survey of the population of a certain country in 1990. The population was divided into three age groups: 0-14, 15-64, and 65 and over. The population of each age group is given in the table below.



1. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the x -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in length ΔL of the block.

SOLUTION: The change in length ΔL can be determined using Hooke's Law:

$$\Delta L = \frac{\sigma L}{E}$$

$$\Delta L = \frac{\sigma L}{E}$$

$$\Delta L = \frac{\sigma L}{E}$$

$$\Delta L = \frac{\sigma L}{E}$$

$$\Delta L = \frac{\sigma L}{E}$$



2. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the y -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in width Δb of the block.

SOLUTION: The change in width Δb can be determined using Hooke's Law:

$\Delta b = \frac{\sigma b}{E}$



1. The first part of the report is a summary of the work done during the last year. It is a brief statement of the results of the work, and is intended to give a general idea of the progress made.

2. The second part of the report is a detailed account of the work done during the last year. It is a full and complete statement of the results of the work, and is intended to give a detailed account of the progress made.

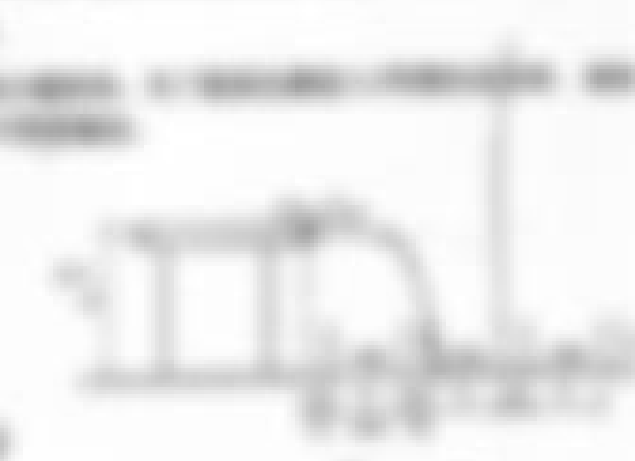


3. The third part of the report is a summary of the work done during the last year. It is a brief statement of the results of the work, and is intended to give a general idea of the progress made.

4. The fourth part of the report is a detailed account of the work done during the last year. It is a full and complete statement of the results of the work, and is intended to give a detailed account of the progress made.

5. The fifth part of the report is a summary of the work done during the last year. It is a brief statement of the results of the work, and is intended to give a general idea of the progress made.

6. The sixth part of the report is a detailed account of the work done during the last year. It is a full and complete statement of the results of the work, and is intended to give a detailed account of the progress made.



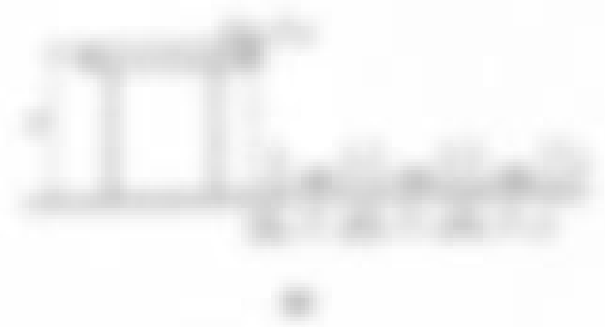
1. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/61. The aircraft was flying at an altitude of 10,000 feet and was in a steep climb when it crashed into the ground.

2. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/61.



3. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/61. The aircraft was flying at an altitude of 10,000 feet and was in a steep climb when it crashed into the ground.

4. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/61. The aircraft was flying at an altitude of 10,000 feet and was in a steep climb when it crashed into the ground.



1. [Illegible text]

2. [Illegible text]

1/2

$\frac{1}{2} \times 10 = 5$

$\frac{1}{2} \times 20 = 10$

$\frac{1}{2} \times 30 = 15$

$\frac{1}{2} \times 40 = 20$

$\frac{1}{2} \times 50 = 25$



Diagram showing a rectangular structure with dimensions and labels.

3. [Illegible text]

4. [Illegible text]

1/2

$\frac{1}{2} \times 10 = 5$

$\frac{1}{2} \times 20 = 10$

$\frac{1}{2} \times 30 = 15$

$\frac{1}{2} \times 40 = 20$

$\frac{1}{2} \times 50 = 25$

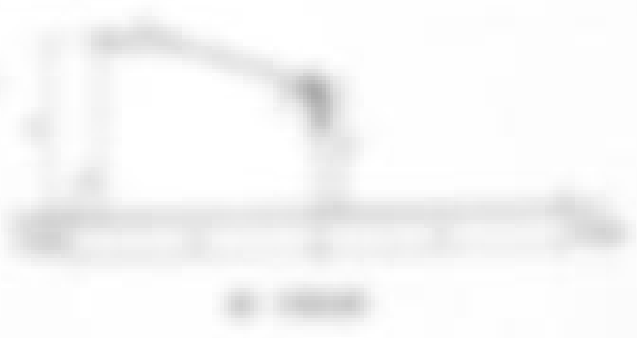


Diagram showing a rectangular structure with dimensions and labels.

1. The following information was obtained from a survey of 100 students regarding their favorite subject in school. The results are shown in the table below.

Subject Number of Students

Math 35
Science 25
History 15
Literature 10
Art 5
Physical Education 5
Music 5



2. The following information was obtained from a survey of 100 students regarding their favorite sport. The results are shown in the table below.

Sport Number of Students

3. The following information was obtained from a survey of 100 students regarding their favorite color. The results are shown in the table below.

Baseball 30
Soccer 20
Basketball 15
Tennis 10
Golf 5
Hockey 5
Other 5



4. The following information was obtained from a survey of 100 students regarding their favorite food. The results are shown in the table below.

1. The following information is for your information only. It is not to be used for any other purpose.

RESEARCH INFORMATION SYSTEMS

1.1.1. The following information is for your information only.

1.1.2. The following information is for your information only.

1.1.3. The following information is for your information only.

1.1.4. The following information is for your information only.

1.1.5. The following information is for your information only.

1.1.6. The following information is for your information only.

1.1.7. The following information is for your information only.

1.1.8. The following information is for your information only.

2. The following information is for your information only. It is not to be used for any other purpose.

RESEARCH INFORMATION SYSTEMS

2.1. The following information is for your information only. It is not to be used for any other purpose.



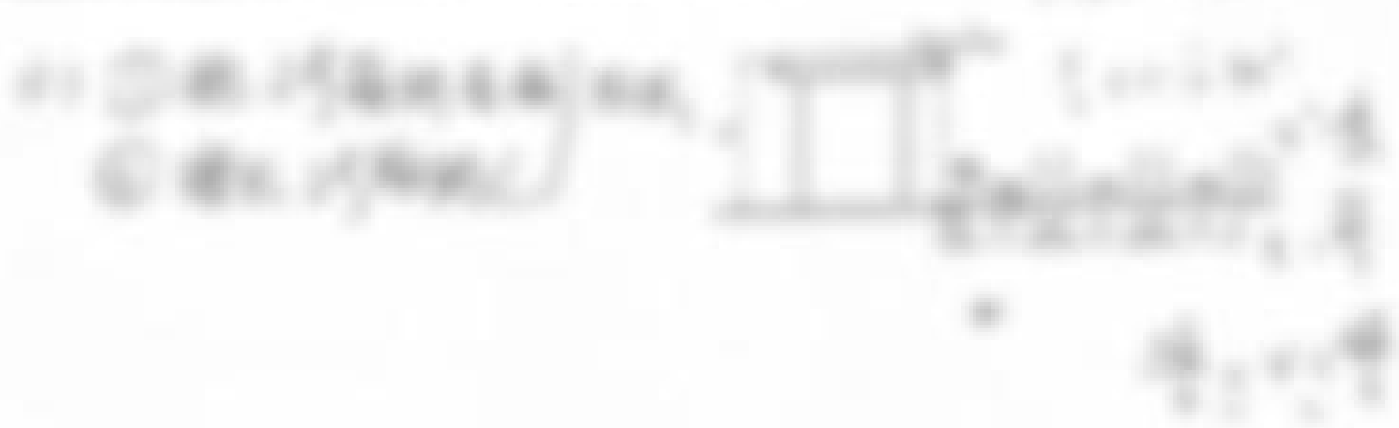
1. 圖中所示之剛架，其左端為固定端，右端為自由端。剛架之總長為 10 公尺，其中間有一垂直之柱，柱高為 4 公尺。剛架之重量為 100 公斤，作用於柱頂。求剛架之反力。

圖中所示之剛架，其左端為固定端，右端為自由端。剛架之總長為 10 公尺，其中間有一垂直之柱，柱高為 4 公尺。剛架之重量為 100 公斤，作用於柱頂。求剛架之反力。



2. 圖中所示之剛架，其左端為固定端，右端為自由端。剛架之總長為 10 公尺，其中間有一垂直之柱，柱高為 4 公尺。剛架之重量為 100 公斤，作用於柱頂。求剛架之反力。

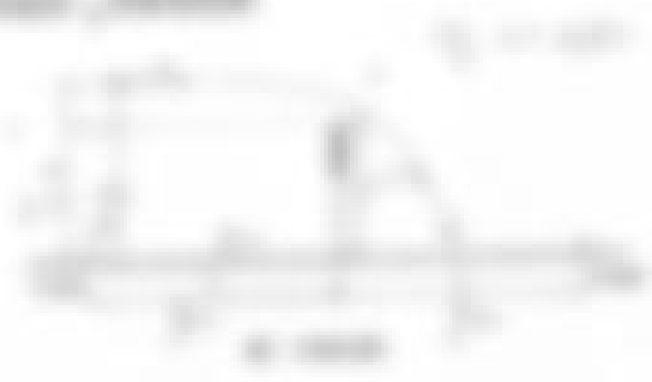
圖中所示之剛架，其左端為固定端，右端為自由端。剛架之總長為 10 公尺，其中間有一垂直之柱，柱高為 4 公尺。剛架之重量為 100 公斤，作用於柱頂。求剛架之反力。



1. **PROBLEM STATEMENT:** A particle of mass m moves in a circular path of radius r with a constant speed v . Find the magnitude of the centripetal force acting on the particle.

SOLUTION: Centripetal force is given by the formula:

$$F_c = \frac{mv^2}{r}$$



$$F_c = \frac{mv^2}{r}$$

2. **PROBLEM STATEMENT:** A particle of mass m moves in a circular path of radius r with a constant speed v . Find the magnitude of the centripetal force acting on the particle.

SOLUTION: Centripetal force is given by the formula:

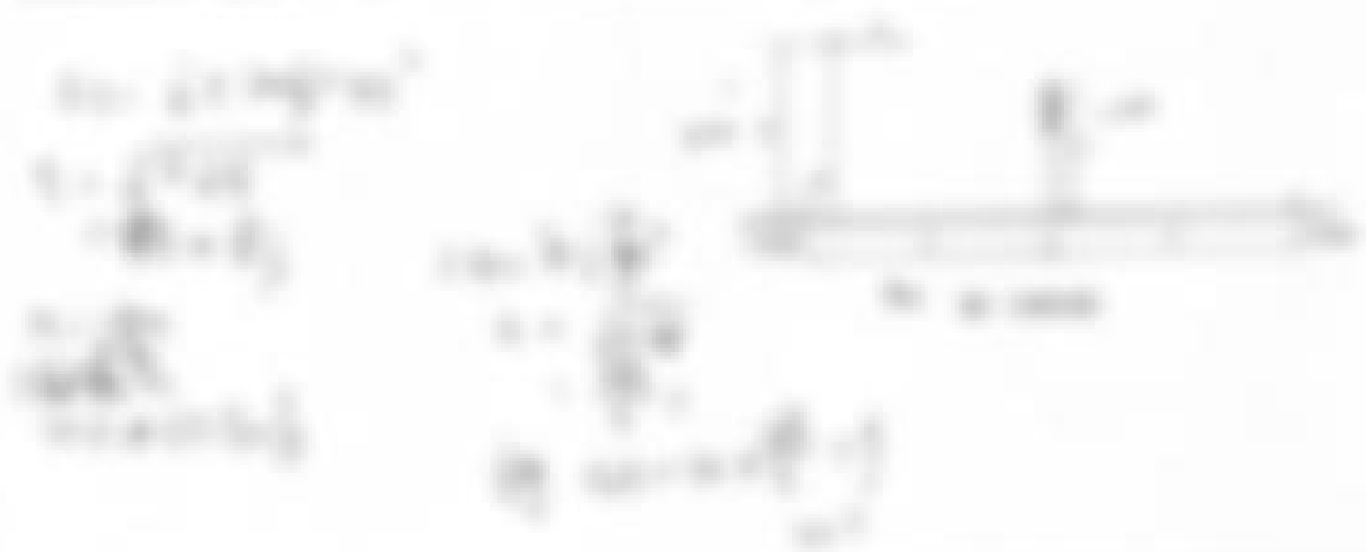


$$F_c = \frac{mv^2}{r}$$

4

1. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.

2. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.



3. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.

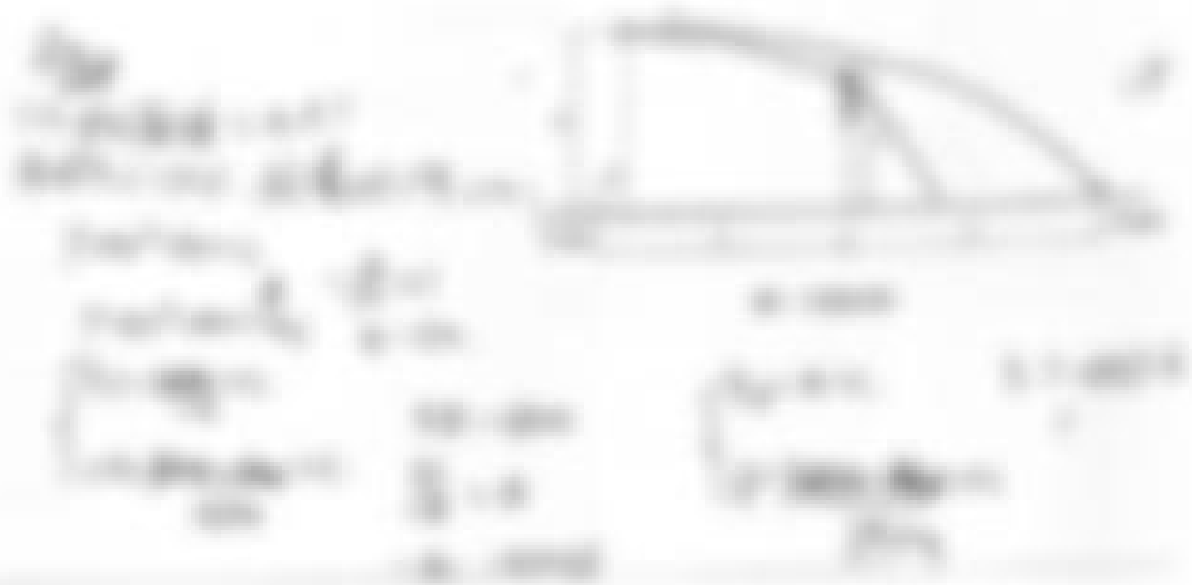
4. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.



5. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000. The following information was obtained from a survey of the population of a certain city. The population of the city is 100,000.

1. **CONSTRUCTION OF THE**
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CONSTRUCTION OF THE PROBLEM. STATEMENT OF THE PROBLEM.



2. **ANALYSIS OF THE PROBLEM. STATEMENT OF THE PROBLEM. STATEMENT OF THE PROBLEM.**
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3. **CONSTRUCTION OF THE PROBLEM. STATEMENT OF THE PROBLEM.**
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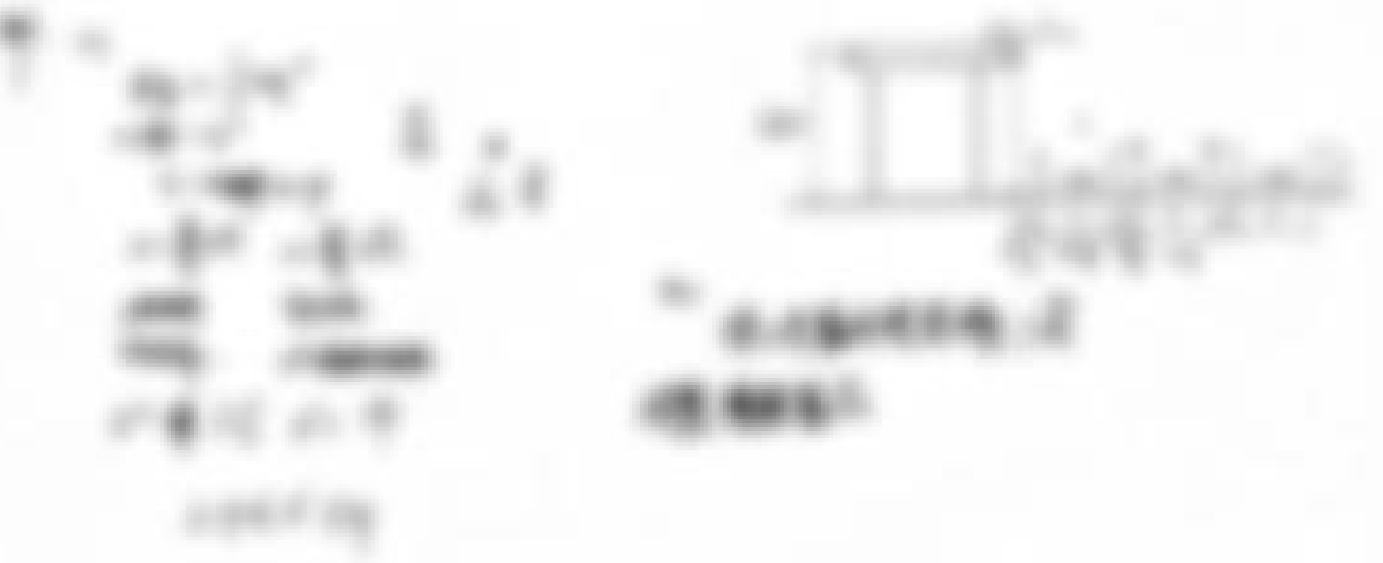


1. [Illegible text]



2. [Illegible text]

3. [Illegible text]



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1. **Abstract:** This paper presents a new method for the automatic detection of the onset of a seizure. The method is based on the analysis of the non-linear properties of the EEG signal. The results show that the proposed method is able to detect the onset of a seizure with a high accuracy and a low false alarm rate.

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1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional study. The data were collected from a convenience sample of 100 nurses working in the intensive care unit (ICU) of a tertiary care hospital in the United States. The sample was selected using a purposive sampling technique to ensure that the participants had the necessary knowledge and experience to provide meaningful responses to the questionnaire. The data were collected over a period of 6 months.

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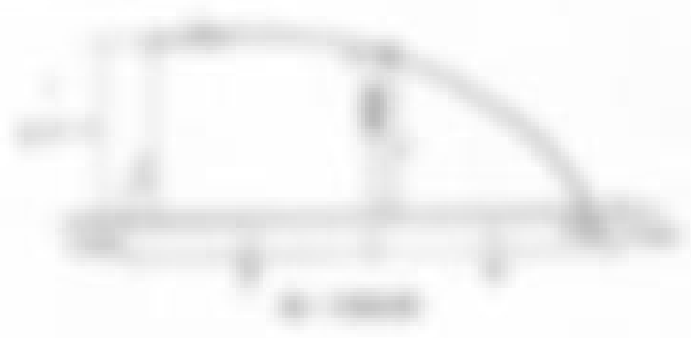
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1. **Abstract.** *Abstract text describing the paper's content.*

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2. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/76.

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4. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/76. The aircraft was a Cessna 441Q, registered N12345, and was operated by the owner, Mr. John Doe. The aircraft was flying at an altitude of 1,000 feet above the ground when it crashed into a field near the town of Smithville, Tennessee. The crash site was located approximately 2 miles south of the town.

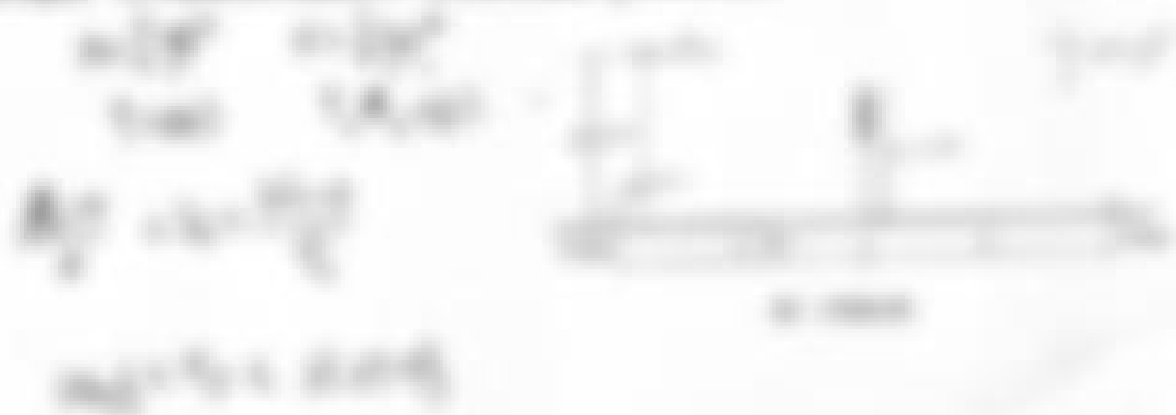
5. The following information was obtained from the investigation of the accident involving the aircraft on 10/10/76.

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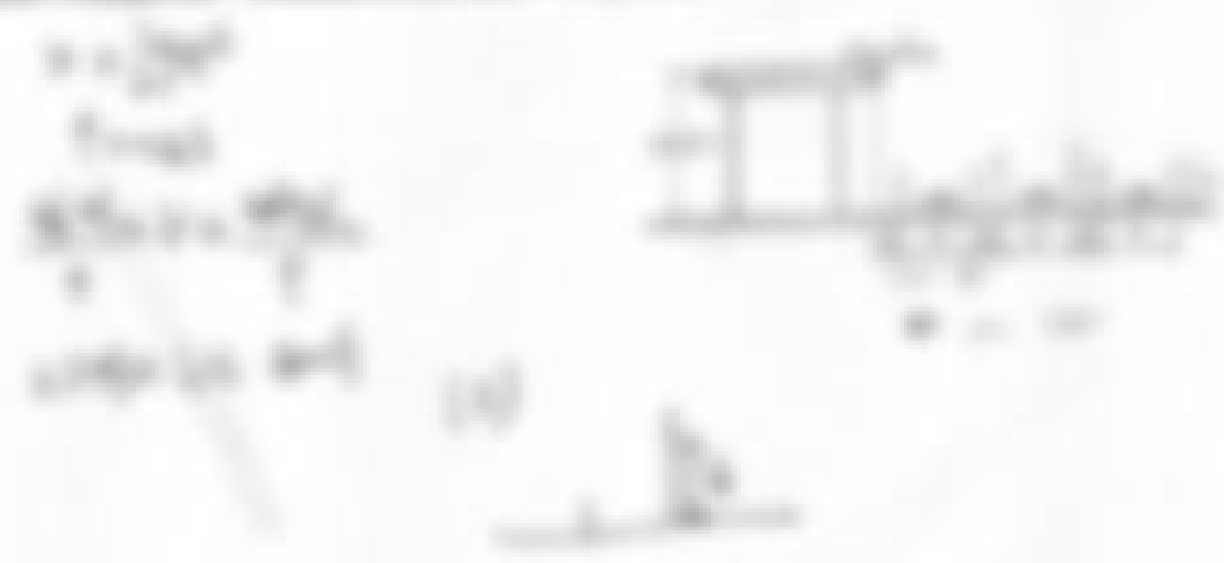
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2. [Illegible text]

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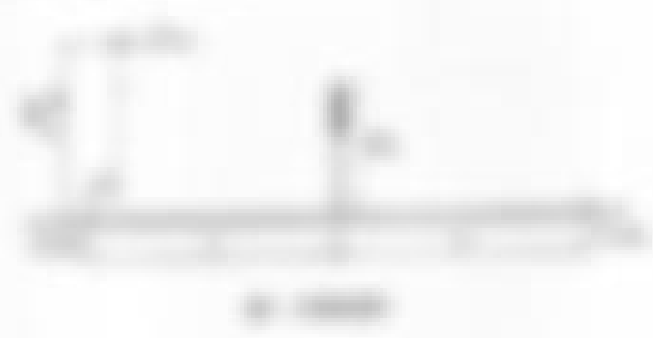
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1. The following information was obtained from the investigation of the accident on 10/10/77. The aircraft was flying at an altitude of 10,000 feet and was in a climb when it encountered a clear air turbulence (CAT) area. The aircraft was damaged and the crew was injured.

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4. The following information was obtained from the investigation of the accident on 10/10/77.

$$P = \frac{1}{2} \rho v^2 C_d A$$

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- 6. The following information was obtained from the investigation of the accident on 10/10/77.
- 7. The following information was obtained from the investigation of the accident on 10/10/77.

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$$P = \frac{1}{2} \rho v^2 C_d A$$

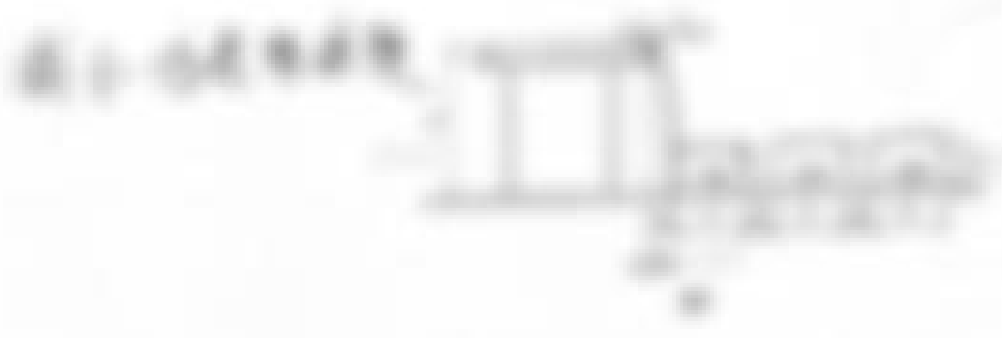
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1. **QUESTION** (10 marks): A particle of mass m moves in a circular path of radius r with a constant speed v . Calculate the centripetal acceleration and the centripetal force acting on the particle.

ANSWER (10 marks): Centripetal acceleration $a_c = \frac{v^2}{r}$

Given:
 $m = 2 \text{ kg}$
 $r = 1 \text{ m}$
 $v = 4 \text{ m/s}$



Find:
1. Centripetal acceleration
2. Centripetal force



2. **QUESTION** (10 marks): A particle of mass m moves in a circular path of radius r with a constant speed v . Calculate the centripetal acceleration and the centripetal force acting on the particle.

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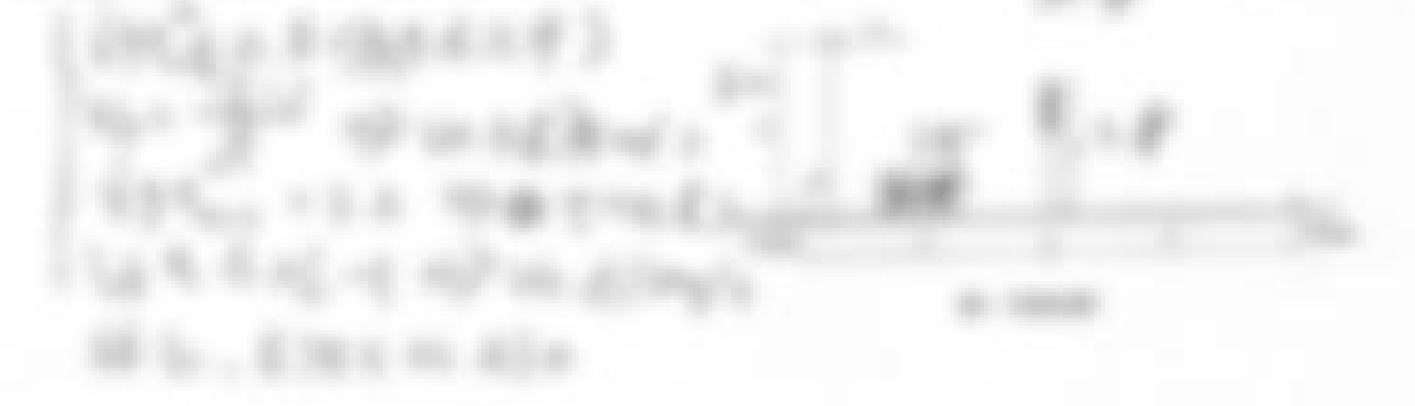
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Find:
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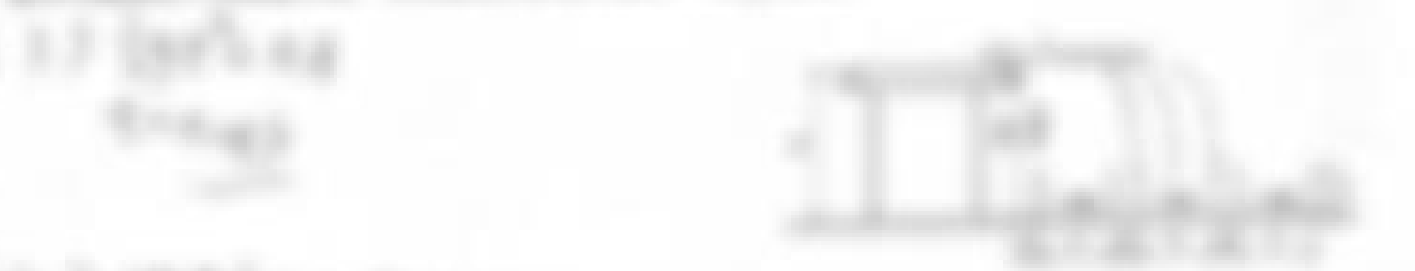


1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top edge of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .



2. A uniform rod of length L and mass M is pivoted at one end. A force F is applied to the other end, perpendicular to the rod. The rod is in equilibrium. Find the magnitude of the force F .

- a. $F = \frac{MgL}{2}$
- b. $F = \frac{MgL}{3}$
- c. $F = \frac{MgL}{4}$
- d. $F = \frac{MgL}{5}$



3. A uniform rod of length L and mass M is pivoted at one end. A force F is applied to the other end, at an angle θ to the rod. The rod is in equilibrium. Find the magnitude of the force F .

1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The vertical displacement of the top face is denoted by δ .

2. The block is made of a material with a Young's modulus E and a Poisson's ratio ν .



3. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The vertical displacement of the top face is denoted by δ .

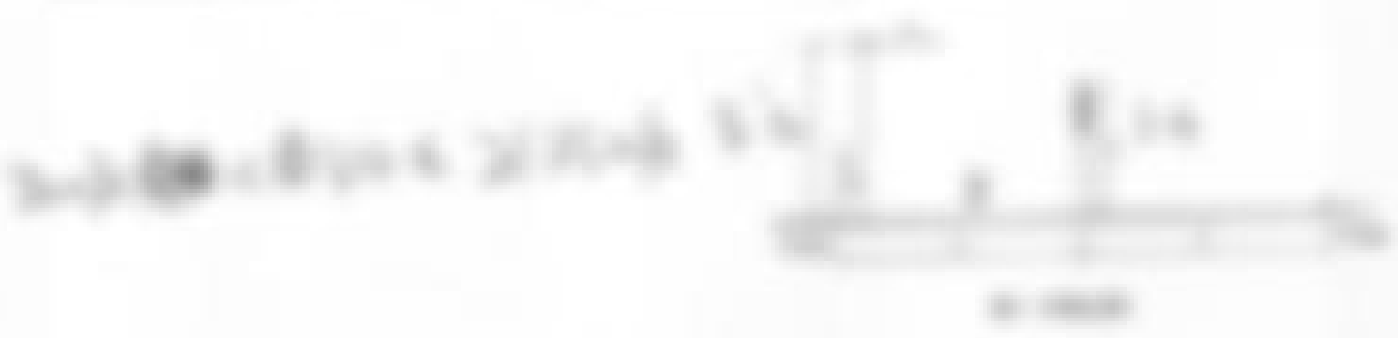
4. The block is made of a material with a Young's modulus E and a Poisson's ratio ν .

5. The block is subjected to a uniform stress σ acting on the vertical faces. The block is free to expand in the horizontal direction. The vertical displacement of the top face is denoted by δ .



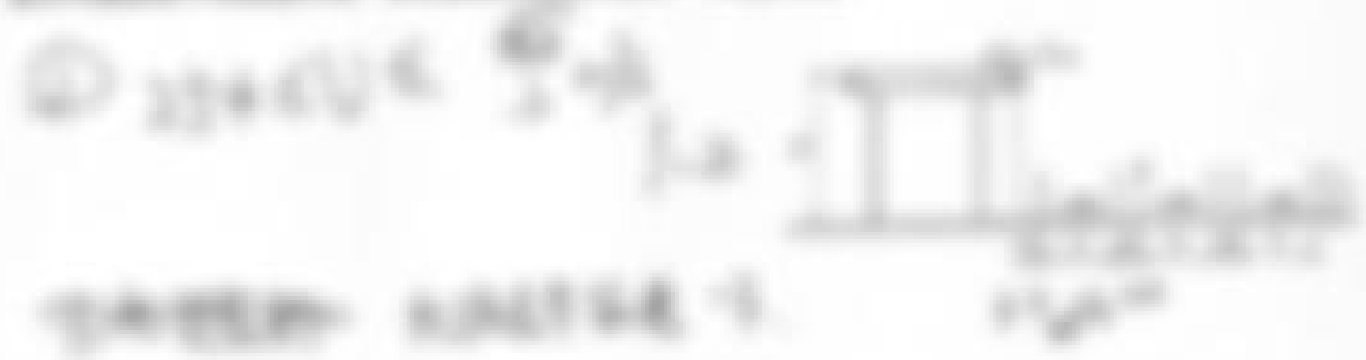
1. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and y .

2. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and y .



3. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and y .

4. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and y .



5. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and y .

1. A rectangular block of mass M and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. The block is on the verge of tipping. Find the magnitude of F .

2. A uniform rod of length L and mass m is pivoted at one end. A force F is applied at the other end, perpendicular to the rod. The rod is in equilibrium. Find the magnitude of F .

• $F = \frac{mg}{2}$

• $F = \frac{mg}{3}$



3. A uniform rod of length L and mass m is pivoted at one end. A force F is applied at the other end, perpendicular to the rod. The rod is in equilibrium. Find the magnitude of F .

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